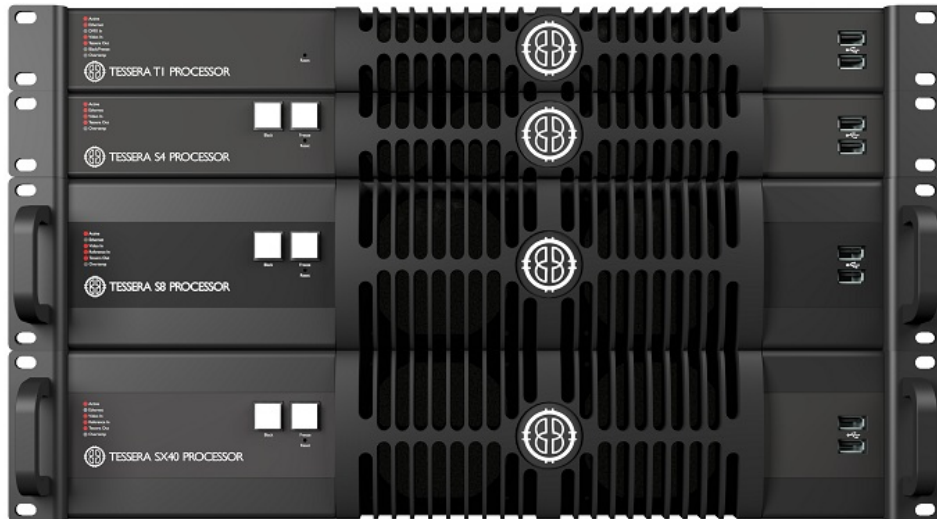


TESSERA PROCESSING IP CONTROL API

SOFTWARE VERSION: 3.5.2

PUBLISH DATE: 20/06/25 18:15



CHANGELOG

3.5.2

- Fixed formatting and precision of floating point values

3.5.0

- Added optional units for numerical endpoints
- Added hidden markers subtree to contain StarTracker endpoints and new custom marker endpoints
- Moved StarTracker endpoints into new hidden markers subtree, marked old paths as deprecated to be removed in a future release
- Added project name endpoint
- Added processor name and serial endpoints
- Added processor temperature sensor endpoints
- Added fan speed and status endpoints
- Added new TestPatternType data type
- Changed test-pattern/type endpoint to accept the TestPatternType data type
- Changed RedundantCableLoopState endpoint's data type from ENUM to STRING and added possible outputs to its description

3.4.0

- Added colour replace endpoints
- Added colour curves endpoints
- Added test pattern restrict to achievable colours switch
- Added group and global brightness overdrive switches
- Added DynaCal highlight out of gamut and overbright pixels switches
- Added Genlock source and internal source framerate endpoints
- Added endpoints for the remaining StarTracker parameters
- Added Dynacal gamut endpoints for all inputs and output
- Removed deprecated input refresh rate and resolution height/width endpoints
- Added custom gradient test pattern endpoints
- Added group and global extended bit depth switches
- Added ShutterSync sensor type and readout time endpoints

3.3.0

- Added StarTracker enabled switch at output/network/startracker/enabled, and group StarTracker enabled and global override switches under group/items/{number}
- Added extensive new input endpoints for each port type under input/ports. input/active/resolution/height and width, and input/active/refresh-rate are now deprecated, to be removed in 3.4
- Added failover trigger endpoint at output/network/failover/actions/request-failover
- Added genlock phase offset controls under output/network/genlock/phase-offset
- Added test pattern custom colour controls under override/test-pattern/custom-colour
- Added system reboot and shutdown trigger endpoints under system/actions

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I. INTRODUCTION

As of version 3.1.0 Tessera processors support remote query and control/triggering functionality over a variety of IP-based protocols via a filesystem-like, RESTful API. This document describes how to access the information available and the format in which it is presented.

The API currently focuses on functionality needed for runtime control of the processor, it does not seek to offer all functions of the processor UI. More functionality will be added in future releases.

In order to use IP control it must be enabled in the Live Control tile in the processor user interface. The processor on the client must be on the same network and have a compatible IP address configuration.

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SUPPORTED PROTOCOLS

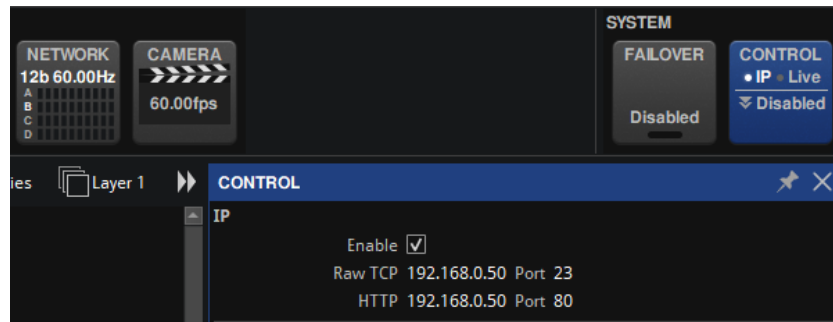
The following protocols are supported by the Tessera API:

- **HTTP** - standard requests via an HTTP client (e.g. web browser)
- **Commandline TCP socket** - Telnet-style text-based commands sent over TCP

Examples of use of each protocol are detailed after the generic control section. All paths, tags and commands are case-insensitive for all protocols.

PROCESSOR SETUP

IP control must be enabled within the project file in order to function, this is done from the control tile in the processor UI. Please refer to the Tessera User Manual for more information.



CONTROL DEVICE SETUP

Make sure that your control device (e.g. laptop) is set to an IP range in the same subnet as the processor, and is physically connected to the same network as your processor's management port. For the purposes of this document, we will be using the factory default IP address of a Brompton processor, 192.168.0.50. Be sure to replace this IP address with that of your processor.

POLLING

Many IP based control systems are set up to 'poll' the target device at fixed intervals to retrieve and update information about that device. Polling the processor has a 'cost' to retrieve the information requested, both for the client (e.g. the laptop sending the API request) and host (e.g. the processor). Consider how often you need to update the information, and the scope of the information you need. Polling large amounts of data multiple times per second may cause adverse performance issues with your processor, so is generally unadvised.

2. API

DATA TYPES

The following endpoint data types are supported:

- **string**: text string of up to 128 UTF-8 encoded characters
- **bool**: boolean state, true or false
- **integer**: signed 16-bit integer, range -32768 to 32767
- **float**: floating point value
- **bytearray**: array of bytes containing binary data
- **enum**: string enumeration representing one of a discrete set of possible values
- **testpattern**: either an enum representing a test pattern, or an integer representing a frame store image by its user number.

COMMANDS

- **get**: get one or more endpoints' value(s)
- **set**: set an endpoint or group of endpoints' value(s)
- **list**: show a summary of available endpoints starting from any position in the API tree
- **help**: show help text for an endpoint or directory detailing what the endpoint represents, access specifier, data format and range

Examples of each command are given in the protocol section.

ACCESS SPECIFIERS

The following access specifiers are supported:

- **R/W**: the endpoint is both readable and writable
- **R/O**: the endpoint is read-only and may not be written to
- **W/O**: the endpoint is write-only and may not be read

ERRORS

The following errors may be returned as the result of a command operation:

- **Path not found**: the requested endpoint path was not recognised
- **Bad operation**: operation was not valid, e.g. trying to set a read-only endpoint
- **Not supported**: not supported by the hardware platform or not yet implemented
- **Missing input parameter**: required input parameters are missing or malformed
- **Bad input parameter type**: one of the input parameters had an incorrect type or format
- **Bad input parameter value**: one of the input parameters had an invalid/out of range value
- **Access denied**: insufficient privilege level for the requested operation
- **No project loaded**: the target processor does not have a project loaded
- **Object not found**: the requested object (e.g. panel) was not found
- **Operation failed**: general runtime failure

DYNAMIC PATHS

Some paths in the API are dynamic in that they depend on the project configuration. For example, accessing group properties depends on what groups have been created in the project. The dynamic sections in the tree are marked with brackets, for example {number}.

3. FULL ENDPOINT REFERENCE

API TREE STRUCTURE

```
api/  
  devices/  
    items/  
      {serial}/  
        firmware  
        type  
      statistics/  
        associated-count  
        error-count  
        online-count  
  groups/  
    items/  
      {number}/  
        brightness  
        brightness-limit/  
          enabled  
          value  
        colour-temperature  
        dark-magic/  
          enabled  
        extended-bit-depth/  
          enabled  
        gains/  
          blue  
          green  
          intensity  
          red  
        gamma  
        global-colour-override  
        global-gains-override  
        global-startracker-override  
        name  
        overdrive/  
          enabled  
        puretone/  
          enabled  
        startracker/  
          enabled  
  input/  
    active/  
      source/  
        port-number  
        port-type  
    ports/  
      dvi/  
        {dvi-port-number}/  
          controls/  
            colour-space/  
              colour  
            dvi-colour-format  
          dynacal/  
            blue/  
              gamut  
              x  
              y  
            green/  
              gamut  
              x
```

- y
- red/
 - gamut
 - x
 - y
- white/
 - colour-temperature
 - gamut
 - x
 - y
- meta-data/
 - refresh-rate
 - resolution/
 - height
 - width
- proc-amp/
 - black-level
 - contrast
 - highlight/
 - blue
 - green
 - red
 - hue
 - saturation
 - shadow/
 - blue
 - green
 - red
- hdmi/
 - {hdmi-port-number}/
 - controls/
 - colour-space/
 - colour
 - info-frame-override-enabled
 - hdmi-colour-format
 - hdr/
 - format
 - pq/
 - auto-brighten
 - gain
 - max-cll-override/
 - enabled
 - luminance
 - quantisation-range
 - dynacal/
 - blue/
 - gamut
 - x
 - y
 - green/
 - gamut
 - x
 - y
 - red/
 - gamut
 - x
 - y
 - white/
 - colour-temperature
 - gamut
 - x
 - y
 - meta-data/
 - bit-depth
 - hdr/
 - format

- refresh-rate
- resolution/
 - height
 - width
- sampling
- proc-amp/
 - black-level
 - contrast
 - highlight/
 - blue
 - green
 - red
 - hue
 - saturation
 - shadow/
 - blue
 - green
 - red
- sdi/
 - {sdi-port-number}/
 - controls/
 - colour-space/
 - colour
 - hdr/
 - format
 - pq/
 - auto-brighten
 - gain
 - max-cll-override/
 - enabled
 - luminance
 - dynacal/
 - blue/
 - gamut
 - x
 - y
 - green/
 - gamut
 - x
 - y
 - red/
 - gamut
 - x
 - y
 - white/
 - colour-temperature
 - gamut
 - x
 - y
 - meta-data/
 - refresh-rate
 - resolution/
 - height
 - width
 - proc-amp/
 - black-level
 - contrast
 - highlight/
 - blue
 - green
 - red
 - hue
 - saturation
 - shadow/
 - blue
 - green

- red
- output/
 - dynacal/
 - {panel-type}/
 - blue/
 - mode
 - x
 - y
 - green/
 - mode
 - x
 - y
 - luminance-only-fraction
 - mode
 - red/
 - mode
 - x
 - y
 - global-colour/
 - brightness
 - brightness-limit/
 - enabled
 - value
 - colour-temperature
 - dark-magic/
 - enabled
 - dynacal/
 - highlight-out-of-gamut-pixels-enabled
 - highlight-overbright-pixels-enabled
 - extended-bit-depth/
 - enabled
 - gains/
 - blue
 - green
 - intensity
 - red
 - gamma
 - overdrive/
 - enabled
 - puretone/
 - enabled
 - network/
 - bit-depth
 - cable-redundancy/
 - loops/
 - {loop-number}/
 - state
 - failover/
 - actions/
 - request-failover
 - settings/
 - enabled
 - modes/
 - on-button-press
 - on-partner-fail
 - on-partner-video-fail
 - prefer-primary
 - role
 - state/
 - is-active
 - is-partner-present
 - partner-absence-duration
 - partner-name
 - partner-serial
 - partner-video-absence-duration
 - frame-rate-multiplier

- frame-remapping/
 - enabled
- frames/
 - {frame}/
 - blue
 - green
 - mode
 - red
 - x-offset
 - y-offset
- genlock/
 - internal-rate
 - phase-offset/
 - absolute/
 - lines
 - pixels
 - angle
 - fraction
 - mode
 - source
- hidden-markers/
 - background-gain
 - blackout-affects-markers
 - closed-shutter-markers
 - custom/
 - data
 - filename
 - marker-gain/
 - blue
 - green
 - red
 - scaling
 - enabled
 - frames-enabled-on
 - mode
- redspy/
 - distance-to-tracker
 - marker-gain
 - marker-seed
 - marker-size-scaler
- startracker/
 - blue/
 - distance-to-tracker
 - enabled
 - marker-gain
 - marker-seed
 - marker-size-scaler
 - green/
 - distance-to-tracker
 - enabled
 - marker-gain
 - marker-seed
 - marker-size-scaler
 - red/
 - distance-to-tracker
 - enabled
 - marker-gain
 - marker-seed
 - marker-size-scaler
- shuttersync/
 - angle-settings/
 - custom-frame-rate
 - shutter-angle
 - use-custom-frame-rate
 - dark-time
 - dark-time-mode

- dark-time-percentage
- insert-dark-time
- mode
- prioritise-refresh-rate
- sensor-readout-time
- sensor-type
- speed-settings/
 - shutter-speed
 - time
- viewer
- startracker/
 - background-gain **(deprecated)**
 - blackout-affects-markers **(deprecated)**
 - blue/
 - distance-to-tracker **(deprecated)**
 - enabled **(deprecated)**
 - marker-gain **(deprecated)**
 - marker-size-scaler **(deprecated)**
 - star-map-seed **(deprecated)**
 - enabled **(deprecated)**
 - frames-enabled-on **(deprecated)**
 - green/
 - distance-to-tracker **(deprecated)**
 - enabled **(deprecated)**
 - marker-gain **(deprecated)**
 - marker-size-scaler **(deprecated)**
 - star-map-seed **(deprecated)**
 - red/
 - distance-to-tracker **(deprecated)**
 - enabled **(deprecated)**
 - marker-gain **(deprecated)**
 - marker-size-scaler **(deprecated)**
 - star-map-seed **(deprecated)**
- override/
 - blackout/
 - enabled
 - fade-time
 - freeze/
 - enabled
 - test-pattern/
 - custom-colour/
 - blue
 - green
 - red
 - custom-gradient/
 - end-colour/
 - blue
 - green
 - red
 - orientation
 - start-colour/
 - blue
 - green
 - red
 - enabled
 - format
 - frame-store/
 - capture-frame
 - delete-frame
 - frames/
 - {frame-user-number}/
 - colour-space
 - enable-alpha
 - format
 - name
 - scaling-mode

- restrict-to-achievable-colours
- type
- presets/
 - active/
 - name
 - number
 - items/
 - {number}/
 - name
 - status
- processing/
 - 3d-lut/
 - data
 - enabled
 - filename
 - strength
 - colour-correct/
 - black/
 - blue
 - green
 - red
 - blue/
 - brightness
 - hue
 - saturation
 - cobalt/
 - brightness
 - hue
 - saturation
 - crimson/
 - brightness
 - hue
 - saturation
 - cyan/
 - brightness
 - hue
 - saturation
 - enabled
 - green/
 - brightness
 - hue
 - saturation
 - lime/
 - brightness
 - hue
 - saturation
 - magenta/
 - brightness
 - hue
 - saturation
 - orange/
 - brightness
 - hue
 - saturation
 - red/
 - brightness
 - hue
 - saturation
 - turquoise/
 - brightness
 - hue
 - saturation
 - violet/
 - brightness
 - hue
 - saturation

- white/
 - blue
 - green
 - red
- yellow/
 - brightness
 - hue
 - saturation
- colour-replace/
 - apply-to-brightness
 - apply-to-hue
 - apply-to-saturation
 - brightness-tolerance
- colour-from/
 - blue
 - green
 - red
- colour-to/
 - blue
 - green
 - red
- colour-tolerance
- enabled
- method
- softness
- strength
- view-matte
- curves/
 - blue/
 - points
 - enabled
 - green/
 - points
 - red/
 - points
 - white/
 - points
- osca/
 - module-correction-enabled
 - seam-correction-enabled
- scaler/
 - enabled
- project/
 - name
- system/
 - actions/
 - reboot
 - shutdown
- current-date-time
- fan/
 - case/
 - one/
 - speed
 - status
 - two/
 - speed
 - status
 - fpga/
 - speed
 - status
- processor-name
- processor-type
- serial-number
- software-version
- temperature/
 - ambient

cpu
dsp
ethernet/
 copper/
 a
 b
 sfp/
 a
 b
 c
 d
fpga
front
gpu
main
psu
rear
uptime

ENDPOINTS DESCRIPTION

DEVICES

DEVICE FIRMWARE

Path: **devices/items/{serial}/firmware**

Description: Current device firmware version

Data type: string

Access Specifier: ReadOnly

DEVICE TYPE

Path: **devices/items/{serial}/type**

Description: Device type name

Data type: string

Access Specifier: ReadOnly

ASSOCIATED DEVICES COUNT

Path: **devices/statistics/associated-count**

Description: The number of devices currently being controlled by the processor

Data type: int

Range: 0 - 2200

Access Specifier: ReadOnly

ERROR DEVICES COUNT

Path: **devices/statistics/error-count**

Description: The number of online devices currently reporting an error state

Data type: int

Range: 0 - 2048

Access Specifier: ReadOnly

ONLINE DEVICE COUNT

Path: **devices/statistics/online-count**

Description: The number of online devices currently detected by the processor

Data type: int

Range: 0 - 2048

Access Specifier: ReadOnly

GROUPS

GROUP BRIGHTNESS

Path: **groups/items/{number}/brightness**

Description: Gets or sets the group output brightness/luminance

Data type: int

Units: Nits

Range: 0 - 10000

Access Specifier: ReadWrite

GROUP BRIGHTNESS LIMIT

Path: **groups/items/{number}/brightness-limit/enabled**

Description: Enables or disables group brightness limit

Data type: bool
Access Specifier: ReadOnly

GROUP BRIGHTNESS LIMIT VALUE

Path: **groups/items/{number}/brightness-limit/value**
Description: Current group maximum brightness value if brightness limit enabled
Data type: int
Units: Nits
Range: 0 - 10000
Access Specifier: ReadOnly

GROUP COLOUR TEMPERATURE

Path: **groups/items/{number}/colour-temperature**
Description: Gets or sets the group colour temperature
Data type: int
Units: Kelvin
Range: 2000 - 11000
Access Specifier: ReadWrite

GROUP DARK MAGIC ENABLED

Path: **groups/items/{number}/dark-magic/enabled**
Description: Enables or disables group Dark Magic
Data type: bool
Access Specifier: ReadWrite

GROUP EXTENDED BIT DEPTH

Path: **groups/items/{number}/extended-bit-depth/enabled**
Description: Enables or disables group extended bit depth
Data type: bool
Access Specifier: ReadWrite

GROUP BLUE GAIN

Path: **groups/items/{number}/gains/blue**
Description: Gets or sets the value of the group blue gain
Data type: float
Units: Percentage
Range: 0 - 100
Decimal places: 2
Access Specifier: ReadWrite

GROUP GREEN GAIN

Path: **groups/items/{number}/gains/green**
Description: Gets or sets the value of the group green gain
Data type: float
Units: Percentage
Range: 0 - 100
Decimal places: 2
Access Specifier: ReadWrite

GROUP INTENSITY GAIN

Path: **groups/items/{number}/gains/intensity**
Description: Gets or sets the value of the group intensity gain
Data type: float
Units: Percentage
Range: 0 - 100
Decimal places: 2

Access Specifier: ReadWrite

GROUP RED GAIN

Path: **groups/items/{number}/gains/red**

Description: Gets or sets the value of the group red gain

Data type: float

Units: Percentage

Range: 0 - 100

Decimal places: 2

Access Specifier: ReadWrite

GROUP OUTPUT GAMMA

Path: **groups/items/{number}/gamma**

Description: Gets or sets the group gamma value

Data type: float

Range: 0.2 - 4.0

Decimal places: 2

Access Specifier: ReadWrite

GROUP GLOBAL COLOUR OVERRIDE

Path: **groups/items/{number}/global-colour-override**

Description: Enables or disables group global colour override

Data type: bool

Access Specifier: ReadWrite

GROUP GLOBAL GAINS OVERRIDE

Path: **groups/items/{number}/global-gains-override**

Description: Enables or disables group global gains override

Data type: bool

Access Specifier: ReadWrite

GROUP GLOBAL STARTRACKER OVERRIDE

Path: **groups/items/{number}/global-startracker-override**

Description: Enables or disables global StarTracker override for this group

Data type: bool

Access Specifier: ReadWrite

GROUP NAME

Path: **groups/items/{number}/name**

Description: Gets or sets the group name

Data type: string

Access Specifier: ReadWrite

GROUP OVERDRIVE ENABLED

Path: **groups/items/{number}/overdrive/enabled**

Description: Enables or disables group brightness overdrive

Data type: bool

Access Specifier: ReadWrite

GROUP PURE TONE ENABLED

Path: **groups/items/{number}/puretone/enabled**

Description: Enables or disables group PureTone

Data type: bool

Access Specifier: ReadWrite

GROUP STARTRACKER ENABLED

Path: **groups/items/{number}/startracker/enabled**

Description: Enables or disables markers in group StarTracker override

Data type: bool

Access Specifier: ReadWrite

INPUT

INPUT PORT NUMBER

Path: **input/active/source/port-number**

Description: Which physical port instance is currently enabled for video input. For example, SDI A = port 1, SDI B = port 2. The available number of port instances for any port type will vary based on the processor hardware variant.

Data type: int

Range: 1 - 2

Access Specifier: ReadWrite

INPUT PORT TYPE

Path: **input/active/source/port-type**

Description: Which physical port instance is currently enabled for video input. The available types will vary based on the processor hardware variant.

Data type: enum

Supported values: dvi, hdmi, sdi

Access Specifier: ReadWrite

DVI INPUT COLOUR SPACE

Path: **input/ports/dvi/{dvi-port-number}/controls/colour-space/colour**

Description: Gets or sets the colour space used for the incoming DVI content

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

INPUT DVI COLOUR FORMAT

Path: **input/ports/dvi/{dvi-port-number}/controls/dvi-colour-format**

Description: Gets or sets the colour format applied by the processor to the input for DVI input

Data type: enum

Supported values: rgb, ypbpr601, ypbpr709

Access Specifier: ReadWrite

DVI DYNACAL BLUE GAMUT

Path: **input/ports/dvi/{dvi-port-number}/dynacal/blue/gamut**

Description: Blue component of DVI input colour space

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

DVI DYNACAL BLUE X

Path: **input/ports/dvi/{dvi-port-number}/dynacal/blue/x**

Description: Blue component X value of DVI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL BLUE Y

Path: **input/ports/dvi/{dvi-port-number}/dynacal/blue/y**

Description: DVI component Y value of video input colour space

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL GREEN GAMUT

Path: **input/ports/dvi/{dvi-port-number}/dynacal/green/gamut**

Description: Green component of DVI input colour space

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

DVI DYNACAL GREEN X

Path: **input/ports/dvi/{dvi-port-number}/dynacal/green/x**

Description: Green component X value of DVI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL GREEN Y

Path: **input/ports/dvi/{dvi-port-number}/dynacal/green/y**

Description: Green component Y value of DVI input colour space

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL RED GAMUT

Path: **input/ports/dvi/{dvi-port-number}/dynacal/red/gamut**

Description: Red component of DVI input colour space

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

DVI DYNACAL RED X

Path: **input/ports/dvi/{dvi-port-number}/dynacal/red/x**

Description: Red component X value of DVI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL RED Y

Path: **input/ports/dvi/{dvi-port-number}/dynacal/red/y**

Description: Red component Y value of DVI input colour space

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL WHITE COLOUR TEMPERATURE

Path: **input/ports/dvi/{dvi-port-number}/dynacal/white/colour-temperature**

Description: DVI input white component colour temperature

Data type: int

Units: Kelvin

Range: 2000 - 11000

Access Specifier: ReadWrite

DVI DYNACAL WHITE GAMUT

Path: **input/ports/dvi/{dvi-port-number}/dynacal/white/gamut**

Description: White component of DVI input colour space. Set to 'Colour Temperature' to control this white-only parameter

Data type: enum

Supported values: d65, aces, colour-temperature, custom

Access Specifier: ReadWrite

DVI DYNACAL WHITE X

Path: **input/ports/dvi/{dvi-port-number}/dynacal/white/x**

Description: White component X value of DVI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4

Access Specifier: ReadWrite

DVI DYNACAL WHITE Y

Path: **input/ports/dvi/{dvi-port-number}/dynacal/white/y**

Description: DVI input white component Y value

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

DVI INPUT REFRESH RATE

Path: **input/ports/dvi/{dvi-port-number}/meta-data/refresh-rate**

Description: DVI input refresh rate

Data type: float

Units: Hertz

Range: 24 - 250

Decimal places: 1

Access Specifier: ReadOnly

DVI INPUT RESOLUTION HEIGHT

Path: **input/ports/dvi/{dvi-port-number}/meta-data/resolution/height**

Description: DVI input height

Data type: int

Units: Pixels

Range: 32 - 4095

Access Specifier: ReadOnly

DVI INPUT RESOLUTION WIDTH

Path: **input/ports/dvi/{dvi-port-number}/meta-data/resolution/width**

Description: DVI input width

Data type: int

Units: Pixels

Range: 32 - 4096

Access Specifier: ReadOnly

DVI INPUT BLACK LEVEL

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/black-level**

Description: Gets or sets black level of a DVI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI INPUT CONTRAST

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/contrast**

Description: Gets or sets contrast of a DVI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI BLUE HIGHLIGHT

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/highlight/blue**

Description: Gets or sets blue highlight of a DVI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI GREEN HIGHLIGHT

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/highlight/green**

Description: Gets or sets green highlight of a DVI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI RED HIGHLIGHT

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/highlight/red**

Description: Gets or sets red highlight of a DVI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI INPUT HUE

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/hue**

Description: Gets or sets hue of a DVI video input. 0° is the passthrough value

Data type: int

Units: Degrees

Range: -180 - 180

Access Specifier: ReadWrite

DVI INPUT SATURATION

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/saturation**

Description: Gets or sets saturation of a DVI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI BLUE SHADOW

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/shadow/blue**

Description: Gets or sets blue shadow of a DVI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

DVI GREEN SHADOW

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/shadow/green**

Description: Gets or sets green shadow of a DVI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

RED SHADOW

Path: **input/ports/dvi/{dvi-port-number}/proc-amp/shadow/red**

Description: Gets or sets red shadow of a DVI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI INPUT COLOUR SPACE

Path: **input/ports/hdmi/{hdmi-port-number}/controls/colour-space/colour**

Description: Gets or sets the colour space used for the incoming HDMI content

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

INFOFRAME OVERRIDE ENABLED

Path: **input/ports/hdmi/{hdmi-port-number}/controls/colour-space/info-frame-override-enabled**

Description: Enables or disables InfoFrame override

Data type: bool

Access Specifier: ReadWrite

HDMI INPUT COLOUR FORMAT

Path: **input/ports/hdmi/{hdmi-port-number}/controls/hdmi-colour-format**

Description: Gets or sets the colour format applied by the processor to the input for HDMI input

Data type: enum

Supported values: from-input, rgb, ycbcr

Access Specifier: ReadWrite

HDMI INPUT HDR FORMAT OVERRIDE

Path: **input/ports/hdmi/{hdmi-port-number}/controls/hdr/format**

Description: Gets or sets the HDR format applied by the processor to the input for HDMI input

Data type: enum

Supported values: from-input, standard-dynamic-range, perceptual-quantiser, hybrid-log-gamma

Access Specifier: ReadWrite

HDMI PQ AUTO BRIGHTEN

Path: **input/ports/hdmi/{hdmi-port-number}/controls/hdr/pq/auto-brighten**

Description: Enables or disables PQ auto brighten for HDMI input

Data type: bool

Access Specifier: ReadWrite

HDMI PQ GAIN

Path: **input/ports/hdmi/{hdmi-port-number}/controls/hdr/pq/gain**

Description: Gets or sets the value of the PQ gain for HDMI input

Data type: float

Range: 0.1 - 10.0

Decimal places: 2

Access Specifier: ReadWrite

HDMI PQ MAXCLL OVERRIDE ENABLED

Path: **input/ports/hdmi/{hdmi-port-number}/controls/hdr/pq/max-cll-override/enabled**

Description: Enables or disables PQ MaxCLL override for HDMI input

Data type: bool

Access Specifier: ReadWrite

HDMI PQ MAXCLL OVERRIDE LUMINANCE

Path: **input/ports/hdmi/{hdmi-port-number}/controls/hdr/pq/max-cll-override/luminance**

Description: Gets or sets the value of the PQ MaxCLL override luminance for HDMI input

Data type: int

Units: Nits

Range: 1 - 10000

Access Specifier: ReadWrite

HDMI INPUT QUANTISATION RANGE

Path: **input/ports/hdmi/{hdmi-port-number}/controls/quantisation-range**

Description: Gets or sets the quantisation range applied by the processor to the HDMI input

Data type: enum

Supported values: from-input, full, limited

Access Specifier: ReadWrite

HDMI DYNACAL BLUE GAMUT

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/blue/gamut**

Description: Blue component of HDMI input colour space

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

HDMI DYNACAL BLUE X

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/blue/x**

Description: Blue component X value of HDMI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4

Access Specifier: ReadWrite

HDMI DYNACAL BLUE Y

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/blue/y**

Description: Blue component Y value of HDMI input colour space

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

HDMI DYNACAL GREEN GAMUT

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/green/gamut**

Description: Green component of HDMI input colour space

Data type: enum
Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom
Access Specifier: ReadWrite

HDMI DYNACAL GREEN X

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/green/x**
Description: Green component X value of HDMI input colour space
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

HDMI DYNACAL GREEN Y

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/green/y**
Description: Green component Y value of HDMI input colour space
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

HDMI DYNACAL RED GAMUT

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/red/gamut**
Description: Red component of HDMI input colour space
Data type: enum
Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom
Access Specifier: ReadWrite

HDMI DYNACAL RED X

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/red/x**
Description: Red component X value of HDMI input colour space
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

HDMI DYNACAL RED Y

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/red/y**
Description: Red component Y value of HDMI input colour space
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

HDMI DYNACAL WHITE COLOUR TEMPERATURE

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/white/colour-temperature**
Description: HDMI input white component colour temperature
Data type: int
Units: Kelvin
Range: 2000 - 11000
Access Specifier: ReadWrite

HDMI DYNACAL WHITE GAMUT

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/white/gamut**
Description: White component of HDMI input colour space. Set to 'Colour Temperature' to control this white-only parameter
Data type: enum

Supported values: d65, aces, colour-temperature, custom
Access Specifier: ReadWrite

HDMI DYNACAL WHITE X

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/white/x**
Description: White component X value of HDMI input colour space
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

HDMI DYNACAL WHITE Y

Path: **input/ports/hdmi/{hdmi-port-number}/dynacal/white/y**
Description: HDMI input white component Y value
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

HDMI INPUT BIT DEPTH

Path: **input/ports/hdmi/{hdmi-port-number}/meta-data/bit-depth**
Description: Gets bit depth of HDMI video input. Valid values are 8, 10, and 12
Data type: int
Units: Bits
Range: 8 - 12
Access Specifier: ReadOnly

HDMI INPUT HDR FORMAT

Path: **input/ports/hdmi/{hdmi-port-number}/meta-data/hdr/format**
Description: HDR format reported by the HDMI input
Data type: enum
Supported values: standard-dynamic-range, perceptual-quantiser, hybrid-log-gamma
Access Specifier: ReadOnly

HDMI INPUT REFRESH RATE

Path: **input/ports/hdmi/{hdmi-port-number}/meta-data/refresh-rate**
Description: Hdmi input refresh rate
Data type: float
Units: Hertz
Range: 24 - 250
Decimal places: 1
Access Specifier: ReadOnly

HDMI INPUT RESOLUTION HEIGHT

Path: **input/ports/hdmi/{hdmi-port-number}/meta-data/resolution/height**
Description: HDMI input height
Data type: int
Units: Pixels
Range: 32 - 4095
Access Specifier: ReadOnly

HDMI INPUT RESOLUTION WIDTH

Path: **input/ports/hdmi/{hdmi-port-number}/meta-data/resolution/width**
Description: HDMI input width
Data type: int
Units: Pixels

Range: 32 - 4096
Access Specifier: ReadOnly

HDMI INPUT SAMPLING

Path: **input/ports/hdmi/{hdmi-port-number}/meta-data/sampling**

Description: Gets current HDMI sampling scheme

Data type: enum

Supported values: rgb, ycbcr422, ycbcr444, ycbcr420

Access Specifier: ReadOnly

HDMI INPUT BLACK LEVEL

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/black-level**

Description: Gets or sets black level of an HDMI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI INPUT CONTRAST

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/contrast**

Description: Gets or sets contrast of an HDMI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI BLUE HIGHLIGHT

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/highlight/blue**

Description: Gets or sets blue highlight of an HDMI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI GREEN HIGHLIGHT

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/highlight/green**

Description: Gets or sets green highlight of an HDMI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI RED HIGHLIGHT

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/highlight/red**

Description: Gets or sets red highlight of an HDMI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI INPUT HUE

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/hue**

Description: Gets or sets hue of an HDMI video input. 0° is the passthrough value

Data type: int

Units: Degrees

Range: -180 - 180

Access Specifier: ReadWrite

HDMI INPUT SATURATION

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/saturation**

Description: Gets or sets saturation of an HDMI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI BLUE SHADOW

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/shadow/blue**

Description: Gets or sets blue shadow of an HDMI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI GREEN SHADOW

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/shadow/green**

Description: Gets or sets green shadow of an HDMI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

HDMI RED SHADOW

Path: **input/ports/hdmi/{hdmi-port-number}/proc-amp/shadow/red**

Description: Gets or sets red shadow of an HDMI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

SDI INPUT COLOUR SPACE

Path: **input/ports/sdi/{sdi-port-number}/controls/colour-space/colour**

Description: Gets or sets the colour space used for the incoming SDI content

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

SDI INPUT HDR FORMAT OVERRIDE

Path: **input/ports/sdi/{sdi-port-number}/controls/hdr/format**

Description: Gets or sets the HDR format applied by the processor to the input for SDI input

Data type: enum

Supported values: from-input, standard-dynamic-range, perceptual-quantiser, hybrid-log-gamma

Access Specifier: ReadWrite

SDI PQ AUTO BRIGHTEN

Path: **input/ports/sdi/{sdi-port-number}/controls/hdr/pq/auto-brighten**

Description: Enables or disables PQ auto brighten for SDI input

Data type: bool

Access Specifier: ReadWrite

SDI PQ GAIN

Path: **input/ports/sdi/{sdi-port-number}/controls/hdr/pq/gain**

Description: Gets or sets the value of the PQ gain for SDI input

Data type: float

Range: 0.1 - 10.0

Decimal places: 2

Access Specifier: ReadWrite

SDI PQ MAXCLL OVERRIDE ENABLED

Path: **input/ports/sdi/{sdi-port-number}/controls/hdr/pq/max-cll-override/enabled**

Description: Enables or disables PQ MaxCLL override for SDI input

Data type: bool

Access Specifier: ReadWrite

SDI PQ MAXCLL OVERRIDE LUMINANCE

Path: **input/ports/sdi/{sdi-port-number}/controls/hdr/pq/max-cll-override/luminance**

Description: Gets or sets the value of the PQ MaxCLL override luminance for SDI input

Data type: int

Units: Nits

Range: 1 - 10000

Access Specifier: ReadWrite

SDI DYNACAL BLUE GAMUT

Path: **input/ports/sdi/{sdi-port-number}/dynacal/blue/gamut**

Description: Blue component of SDI input colour space

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

SDI DYNACAL BLUE X

Path: **input/ports/sdi/{sdi-port-number}/dynacal/blue/x**

Description: Blue component X value of SDI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4

Access Specifier: ReadWrite

SDI DYNACAL BLUE Y

Path: **input/ports/sdi/{sdi-port-number}/dynacal/blue/y**

Description: Blue component Y value of SDI input colour space

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

SDI DYNACAL GREEN GAMUT

Path: **input/ports/sdi/{sdi-port-number}/dynacal/green/gamut**

Description: Green component of SDI input colour space

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: ReadWrite

SDI DYNACAL GREEN X

Path: **input/ports/sdi/{sdi-port-number}/dynacal/green/x**

Description: Green component X value of SDI input colour space

Data type: float

Range: 0.0 - 0.8

Decimal places: 4
Access Specifier: ReadWrite

SDI DYNACAL GREEN Y

Path: **input/ports/sdi/{sdi-port-number}/dynacal/green/y**
Description: Green component Y value of SDI input colour space
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

SDI DYNACAL RED GAMUT

Path: **input/ports/sdi/{sdi-port-number}/dynacal/red/gamut**
Description: Red component of SDI input colour space
Data type: enum
Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom
Access Specifier: ReadWrite

SDI DYNACAL RED X

Path: **input/ports/sdi/{sdi-port-number}/dynacal/red/x**
Description: Red component X value of SDI input colour space
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

SDI DYNACAL RED Y

Path: **input/ports/sdi/{sdi-port-number}/dynacal/red/y**
Description: Red component Y value of SDI input colour space
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

SDI DYNACAL WHITE COLOUR TEMPERATURE

Path: **input/ports/sdi/{sdi-port-number}/dynacal/white/colour-temperature**
Description: SDI input white component colour temperature
Data type: int
Units: Kelvin
Range: 2000 - 11000
Access Specifier: ReadWrite

SDI DYNACAL WHITE GAMUT

Path: **input/ports/sdi/{sdi-port-number}/dynacal/white/gamut**
Description: White component of SDI input colour space. Set to 'Colour Temperature' to control this white-only parameter
Data type: enum
Supported values: d65, aces, colour-temperature, custom
Access Specifier: ReadWrite

SDI DYNACAL WHITE X

Path: **input/ports/sdi/{sdi-port-number}/dynacal/white/x**
Description: White component X value of SDI input colour space
Data type: float
Range: 0.0 - 0.8
Decimal places: 4

Access Specifier: ReadWrite

SDI DYNACAL WHITE Y

Path: **input/ports/sdi/{sdi-port-number}/dynacal/white/y**

Description: SDI input white component Y value

Data type: float

Range: 0.0 - 0.9

Decimal places: 4

Access Specifier: ReadWrite

SDI INPUT REFRESH RATE

Path: **input/ports/sdi/{sdi-port-number}/meta-data/refresh-rate**

Description: SDI input refresh rate

Data type: float

Units: Hertz

Range: 24 - 250

Decimal places: 1

Access Specifier: ReadOnly

SDI INPUT RESOLUTION HEIGHT

Path: **input/ports/sdi/{sdi-port-number}/meta-data/resolution/height**

Description: SDI input height

Data type: int

Units: Pixels

Range: 32 - 4095

Access Specifier: ReadOnly

SDI INPUT RESOLUTION WIDTH

Path: **input/ports/sdi/{sdi-port-number}/meta-data/resolution/width**

Description: SDI input width

Data type: int

Units: Pixels

Range: 32 - 4096

Access Specifier: ReadOnly

SDI INPUT BLACK LEVEL

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/black-level**

Description: Gets or sets black level of an SDI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

SDI INPUT CONTRAST

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/contrast**

Description: Gets or sets contrast of an SDI video input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200

Access Specifier: ReadWrite

SDI BLUE HIGHLIGHT

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/highlight/blue**

Description: Gets or sets blue highlight of an SDI input. 100% is the passthrough value

Data type: int

Units: Percentage

Range: 0 - 200
Access Specifier: ReadWrite

SDI GREEN HIGHLIGHT

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/highlight/green**
Description: Gets or sets green highlight of an SDI input. 100% is the passthrough value
Data type: int
Units: Percentage
Range: 0 - 200
Access Specifier: ReadWrite

SDI RED HIGHLIGHT

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/highlight/red**
Description: Gets or sets red highlight of an SDI video input. 100% is the passthrough value
Data type: int
Units: Percentage
Range: 0 - 200
Access Specifier: ReadWrite

SDI INPUT HUE

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/hue**
Description: Gets or sets hue of an SDI video input. 0° is the passthrough value
Data type: int
Units: Degrees
Range: -180 - 180
Access Specifier: ReadWrite

SDI INPUT SATURATION

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/saturation**
Description: Gets or sets saturation of an SDI video input. 100% is the passthrough value
Data type: int
Units: Percentage
Range: 0 - 200
Access Specifier: ReadWrite

SDI BLUE SHADOW

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/shadow/blue**
Description: Gets or sets blue shadow of an SDI input. 100% is the passthrough value
Data type: int
Units: Percentage
Range: 0 - 200
Access Specifier: ReadWrite

SDI GREEN SHADOW

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/shadow/green**
Description: Gets or sets green shadow of an SDI input. 100% is the passthrough value
Data type: int
Units: Percentage
Range: 0 - 200
Access Specifier: ReadWrite

SDI RED SHADOW

Path: **input/ports/sdi/{sdi-port-number}/proc-amp/shadow/red**
Description: Gets or sets red shadow of an SDI video input. 100% is the passthrough value
Data type: int
Units: Percentage

Range: 0 - 200
Access Specifier: ReadWrite

OUTPUT

OUTPUT DYNACAL BLUE MODE

Path: **output/dynacal/{panel-type}/blue/mode**
Description: Gets or sets the blue component of panel output colour space
Data type: enum
Supported values: achievable, custom
Access Specifier: ReadWrite

OUTPUT DYNACAL BLUE X

Path: **output/dynacal/{panel-type}/blue/x**
Description: X value of panel output colour space blue component
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

OUTPUT DYNACAL BLUE Y

Path: **output/dynacal/{panel-type}/blue/y**
Description: Y value of panel output colour space blue component
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

OUTPUT DYNACAL GREEN MODE

Path: **output/dynacal/{panel-type}/green/mode**
Description: Gets or sets the green component of panel output colour space
Data type: enum
Supported values: achievable, custom
Access Specifier: ReadWrite

OUTPUT DYNACAL GREEN X

Path: **output/dynacal/{panel-type}/green/x**
Description: X value of panel output colour space green component
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

OUTPUT DYNACAL GREEN Y

Path: **output/dynacal/{panel-type}/green/y**
Description: Y value of panel output colour space green component
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

OUTPUT DYNACAL LUMINANCE ONLY FRACTION

Path: **output/dynacal/{panel-type}/luminance-only-fraction**

Description: Gets or sets the percentage mix of output calibration
Data type: int
Units: Percentage
Range: 0 - 100
Access Specifier: ReadWrite

OUTPUT DYNACAL MODE

Path: **output/dynacal/{panel-type}/mode**
Description: Gets or sets the panel output colour space
Data type: enum
Supported values: match-input, achievable, custom
Access Specifier: ReadWrite

OUTPUT DYNACAL RED MODE

Path: **output/dynacal/{panel-type}/red/mode**
Description: Gets or sets the red component of panel output colour space
Data type: enum
Supported values: achievable, custom
Access Specifier: ReadWrite

OUTPUT DYNACAL RED X

Path: **output/dynacal/{panel-type}/red/x**
Description: X value of panel output colour space red component
Data type: float
Range: 0.0 - 0.8
Decimal places: 4
Access Specifier: ReadWrite

OUTPUT DYNACAL RED Y

Path: **output/dynacal/{panel-type}/red/y**
Description: Y value of panel output colour space red component
Data type: float
Range: 0.0 - 0.9
Decimal places: 4
Access Specifier: ReadWrite

OUTPUT BRIGHTNESS

Path: **output/global-colour/brightness**
Description: Write -1 to reset output brightness to calculated common maximum for available fixtures.
Data type: int
Units: Nits
Range: -1 - 10000
Access Specifier: ReadWrite

BRIGHTNESS LIMIT ENABLED

Path: **output/global-colour/brightness-limit/enabled**
Description: Enables or disables global brightness limit
Data type: bool
Access Specifier: ReadOnly

BRIGHTNESS LIMIT VALUE

Path: **output/global-colour/brightness-limit/value**
Description: Current maximum brightness value if brightness limit enabled
Data type: int
Units: Nits
Range: 0 - 10000

Access Specifier: ReadOnly

OUTPUT COLOUR TEMPERATURE

Path: **output/global-colour/colour-temperature**

Description: Gets or sets the output colour temperature

Data type: int

Units: Kelvin

Range: 2000 - 11000

Access Specifier: ReadWrite

DARK MAGIC ENABLED

Path: **output/global-colour/dark-magic/enabled**

Description: Enables or disables the processor's Dark Magic feature

Data type: bool

Access Specifier: ReadWrite

HIGHLIGHT OUT OF GAMUT PIXELS ENABLED

Path: **output/global-colour/dynacal/highlight-out-of-gamut-pixels-enabled**

Description: Enables or disables DynaCal's out-of-gamut pixel feature

Data type: bool

Access Specifier: ReadWrite

HIGHLIGHT OVERBRIGHT PIXELS ENABLED

Path: **output/global-colour/dynacal/highlight-overbright-pixels-enabled**

Description: Enables or disables DynaCal's overbright pixel feature

Data type: bool

Access Specifier: ReadWrite

EXTENDED BIT DEPTH ENABLED

Path: **output/global-colour/extended-bit-depth/enabled**

Description: Enables or disables extended bit depth

Data type: bool

Access Specifier: ReadWrite

BLUE GAIN

Path: **output/global-colour/gains/blue**

Description: Gets or sets the value of the output blue gain

Data type: float

Units: Percentage

Range: 0 - 100

Decimal places: 2

Access Specifier: ReadWrite

GREEN GAIN

Path: **output/global-colour/gains/green**

Description: Gets or sets the value of the output green gain

Data type: float

Units: Percentage

Range: 0 - 100

Decimal places: 2

Access Specifier: ReadWrite

INTENSITY GAIN

Path: **output/global-colour/gains/intensity**

Description: Gets or sets the value of the output intensity gain
Data type: float
Units: Percentage
Range: 0 - 100
Decimal places: 2
Access Specifier: ReadWrite

RED GAIN

Path: **output/global-colour/gains/red**
Description: Gets or sets the value of the output red gain
Data type: float
Units: Percentage
Range: 0 - 100
Decimal places: 2
Access Specifier: ReadWrite

OUTPUT GAMMA

Path: **output/global-colour/gamma**
Description: Gets or sets the value of the output gamma
Data type: float
Range: 0.2 - 4.0
Decimal places: 2
Access Specifier: ReadWrite

OVERDRIVE ENABLED

Path: **output/global-colour/overdrive/enabled**
Description: Enables or disables global brightness overdrive
Data type: bool
Access Specifier: ReadWrite

PURE TONE ENABLED

Path: **output/global-colour/puretone/enabled**
Description: Enables or disables PureTone
Data type: bool
Access Specifier: ReadWrite

NETWORK BIT DEPTH

Path: **output/network/bit-depth**
Description: Gets or sets bit depth of video output. Valid values are 8, 10, and 12
Data type: int
Units: Bits
Range: 8 - 12
Access Specifier: ReadWrite

REDUNDANT CABLE LOOP STATE

Path: **output/network/cable-redundancy/loops/{loop-number}/state**
Description: Current state of cable loop redundancy on the processor in the format : ->, where can be: 'loop-found', 'no-loop-found', 'incorrect-loop-found', or 'one-to-many-error' . is the connected trunk followed by the XD port number (e.g. 'A1'). If is 'no-loop-found', can be 'No connection'.
Data type: string
Access Specifier: ReadOnly

REQUEST FAILOVER

Path: **output/network/failover/actions/request-failover**
Description: Send an empty string to activate processor redundancy
Data type: string

Access Specifier: WriteOnly

FAILOVER ENABLED

Path: **output/network/failover/settings/enabled**

Description: Enables or disables failover mode on the processor

Data type: bool

Access Specifier: ReadWrite

BUTTON PRESS FAILOVER MODE ENABLED

Path: **output/network/failover/settings/modes/on-button-press**

Description: Enables or disables failover to backup processor when the processor's Blackout/Freeze buttons are pushed

Data type: bool

Access Specifier: ReadWrite

PARTNER FAILOVER MODE ENABLED

Path: **output/network/failover/settings/modes/on-partner-fail**

Description: Enables or disables partner processor failover when processor failure is detected (e.g. the processor loses power)

Data type: bool

Access Specifier: ReadWrite

PARTNER VIDEO FAILOVER MODE ENABLED

Path: **output/network/failover/settings/modes/on-partner-video-fail**

Description: Enables or disables partner processor failover on video signal loss

Data type: bool

Access Specifier: ReadWrite

PREFER PRIMARY FAILOVER MODE ENABLED

Path: **output/network/failover/settings/modes/prefer-primary**

Description: If prefer primary processor failover mode is activated, when primary processor is functioning correctly, it will be automatically always be the active processor

Data type: bool

Access Specifier: ReadWrite

FAILOVER ROLE

Path: **output/network/failover/settings/role**

Description: Is processor's failover role Primary or Backup

Data type: enum

Supported values: primary, backup

Access Specifier: ReadOnly

FAILOVER IS ACTIVE

Path: **output/network/failover/state/is-active**

Description: Whether failover is active on the processor

Data type: bool

Access Specifier: ReadOnly

FAILOVER PARTNER IS ONLINE

Path: **output/network/failover/state/is-partner-present**

Description: Whether the backup processor is currently online and detected

Data type: bool

Access Specifier: ReadOnly

FAILOVER PARTNER ABSENCE DURATION

Path: **output/network/failover/state/partner-absence-duration**

Description: How long the backup processor has been absent for

Data type: string

Access Specifier: ReadOnly

FAILOVER PARTNER NAME

Path: **output/network/failover/state/partner-name**

Description: Name of the backup processor

Data type: string

Access Specifier: ReadOnly

FAILOVER PARTNER SERIAL

Path: **output/network/failover/state/partner-serial**

Description: Serial number of the backup processor

Data type: string

Access Specifier: ReadOnly

FAILOVER PARTNER VIDEO ABSENCE DURATION

Path: **output/network/failover/state/partner-video-absence-duration**

Description: Time since backup processor video source was last detected

Data type: string

Access Specifier: ReadOnly

NETWORK FRAME RATE MULTIPLIER

Path: **output/network/frame-rate-multiplier**

Description: Gets or sets frame rate multiplier of video output. Set value to 1 to disable frame rate multiplication.

Data type: int

Range: 1 - 10

Access Specifier: ReadWrite

FRAME REMAPPING ENABLED

Path: **output/network/frame-remapping/enabled**

Description: Is frame remapping globally enabled or disabled

Data type: bool

Access Specifier: ReadWrite

FRAME REMAPPING BLUE

Path: **output/network/frame-remapping/frames/{frame}/blue**

Description: Gets or sets blue value of frame colour

Data type: int

Range: 0 - 255

Access Specifier: ReadWrite

FRAME REMAPPING GREEN

Path: **output/network/frame-remapping/frames/{frame}/green**

Description: Gets or sets green value of frame colour

Data type: int

Range: 0 - 255

Access Specifier: ReadWrite

FRAME REMAPPING MODE

Path: **output/network/frame-remapping/frames/{frame}/mode**
Description: Is frame mode Colour or Video
Data type: enum
Supported values: colour, video
Access Specifier: ReadWrite

FRAME REMAPPING RED

Path: **output/network/frame-remapping/frames/{frame}/red**
Description: Gets or sets red value of frame colour
Data type: int
Range: 0 - 255
Access Specifier: ReadWrite

FRAME REMAPPING X OFFSET

Path: **output/network/frame-remapping/frames/{frame}/x-offset**
Description: Gets or sets x offset of frame
Data type: int
Units: Pixels
Range: -4095 - 4095
Access Specifier: ReadWrite

FRAME REMAPPING Y OFFSET

Path: **output/network/frame-remapping/frames/{frame}/y-offset**
Description: Gets or sets y offset of frame
Data type: int
Units: Pixels
Range: -4095 - 4095
Access Specifier: ReadWrite

GENLOCKINTERNALSOURCE RATE

Path: **output/network/genlock/internal-rate**
Description: Gets or sets current refresh rate for internal genlock source
Data type: float
Units: Hertz
Range: 23.5 - 251.0
Decimal places: 4
Access Specifier: ReadWrite

GENLOCK PHASE OFFSET LINES

Path: **output/network/genlock/phase-offset/absolute/lines**
Description: Gets or sets genlock phase offset absolute lines value for the processor
Data type: int
Range: -9999 - 9999
Access Specifier: ReadWrite

GENLOCK PHASE OFFSET PIXELS

Path: **output/network/genlock/phase-offset/absolute/pixels**
Description: Gets or sets genlock phase offset absolute pixels value for the processor
Data type: int
Range: -9999 - 9999
Access Specifier: ReadWrite

GENLOCK PHASE OFFSET ANGLE

Path: **output/network/genlock/phase-offset/angle**
Description: Gets or sets genlock phase offset angle for the processor
Data type: float

Units: Degrees
Range: -360 - 360
Decimal places: 6
Access Specifier: ReadWrite

GENLOCK PHASE OFFSET FRACTION

Path: **output/network/genlock/phase-offset/fraction**
Description: Gets or sets genlock phase offset fraction for the processor
Data type: float
Units: Percentage
Range: -100 - 100
Decimal places: 5
Access Specifier: ReadWrite

GENLOCK PHASE OFFSET MODE

Path: **output/network/genlock/phase-offset/mode**
Description: Gets or sets genlock phase offset mode for the processor
Data type: enum
Supported values: none, angle, fraction, absolute
Access Specifier: ReadWrite

GENLOCK SOURCE

Path: **output/network/genlock/source**
Description: Gets or sets the current genlock source for the processor
Data type: enum
Supported values: internal, sdi, sdi-a, sdi-b, hdmi, dvi, ref-in, active-input
Access Specifier: ReadWrite

HIDDENMARKERS BACKGROUND GAIN

Path: **output/network/hidden-markers/background-gain**
Description: Gets or sets brightness of the video underneath hidden markers as a percentage
Data type: int
Units: Percentage
Range: 0 - 100
Access Specifier: ReadWrite

HIDDENMARKERS BLACKOUT AFFECTS MARKERS

Path: **output/network/hidden-markers/blackout-affects-markers**
Description: Gets or sets whether blackout affects markers as well as video
Data type: bool
Access Specifier: ReadWrite

CLOSED SHUTTER MARKERS

Path: **output/network/hidden-markers/closed-shutter-markers**
Description: Gets or sets whether markers are only shown when camera shutter is closed
Data type: bool
Access Specifier: ReadWrite

CUSTOM MARKERS DATA

Path: **output/network/hidden-markers/custom/data**
Description: Send bulk data in image file format to upload custom markers
Data type: bytearray
Access Specifier: ReadWrite

CUSTOM MARKERS FILENAME

Path: **output/network/hidden-markers/custom/filename**

Description: The name of the stored custom markers file

Data type: string

Access Specifier: ReadOnly

CUSTOM MARKERS BLUE GAIN

Path: **output/network/hidden-markers/custom/marker-gain/blue**

Description: Gets or sets the current brightness of the blue custom markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

CUSTOM MARKERS GREEN GAIN

Path: **output/network/hidden-markers/custom/marker-gain/green**

Description: Gets or sets the current brightness of the green custom markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

CUSTOM MARKERS RED GAIN

Path: **output/network/hidden-markers/custom/marker-gain/red**

Description: Gets or sets the current brightness of the red custom markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

CUSTOM MARKERS SCALING

Path: **output/network/hidden-markers/custom/scaling**

Description: Gets or sets current scaling for hidden markers

Data type: enum

Supported values: 1:1, stretch, fit, fill

Access Specifier: ReadWrite

HIDDENMARKERS ENABLED

Path: **output/network/hidden-markers/enabled**

Description: Enables or disables hidden markers

Data type: bool

Access Specifier: ReadWrite

HIDDENMARKERS FRAMES ENABLED ON

Path: **output/network/hidden-markers/frames-enabled-on**

Description: A valid array must contain ints in the range between 1 and the max supported frame rate multiplier for the platform

Data type: array

Access Specifier: ReadWrite

HIDDEN MARKERS MODE

Path: **output/network/hidden-markers/mode**

Description: Gets or sets current hidden markers mode

Data type: enum

Supported values: none, redspy, startracker, custom

Access Specifier: ReadWrite

REDSPY DISTANCE TO TRACKER

Path: **output/network/hidden-markers/redspy/distance-to-tracker**

Description: Gets or sets distance to tracker for RedSpy markers in metres

Data type: float

Units: Metres

Range: 0.1 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

REDSPY MARKER GAIN

Path: **output/network/hidden-markers/redspy/marker-gain**

Description: Gets or sets the current brightness of the RedSpy markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

REDSPY MARKER SEED

Path: **output/network/hidden-markers/redspy/marker-seed**

Description: Gets or sets seed value for RedSpy marker positions

Data type: int

Range: 1 - 65535

Access Specifier: ReadWrite

REDSPY MARKER SIZE SCALER

Path: **output/network/hidden-markers/redspy/marker-size-scaler**

Description: Gets or sets the size scaling for RedSpy markers as a percentage

Data type: int

Units: Percentage

Range: 10 - 200

Access Specifier: ReadWrite

STARTRACKER BLUE DISTANCE TO TRACKER

Path: **output/network/hidden-markers/startracker/blue/distance-to-tracker**

Description: Gets or sets distance to tracker for blue StarTracker markers in metres

Data type: float

Units: Metres

Range: 0.1 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

STARTRACKER BLUE ENABLED

Path: **output/network/hidden-markers/startracker/blue/enabled**

Description: Enables or disables blue markers for StarTracker

Data type: bool

Access Specifier: ReadWrite

STARTRACKER BLUE MARKER GAIN

Path: **output/network/hidden-markers/startracker/blue/marker-gain**

Description: Gets or sets the current brightness of the blue StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

STARTRACKER BLUE MARKER SEED

Path: **output/network/hidden-markers/startracker/blue/marker-seed**
Description: Gets or sets seed value for blue StarTracker marker positions
Data type: int
Range: 1 - 65535
Access Specifier: ReadWrite

STARTRACKER BLUE MARKER SIZE SCALER

Path: **output/network/hidden-markers/startracker/blue/marker-size-scaler**
Description: Gets or sets the size scaling for blue StarTracker markers as a percentage
Data type: int
Units: Percentage
Range: 10 - 200
Access Specifier: ReadWrite

STARTRACKER GREEN DISTANCE TO TRACKER

Path: **output/network/hidden-markers/startracker/green/distance-to-tracker**
Description: Gets or sets distance to tracker for green StarTracker markers in metres
Data type: float
Units: Metres
Range: 0.1 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

STARTRACKER GREEN ENABLED

Path: **output/network/hidden-markers/startracker/green/enabled**
Description: Enables or disables green markers for StarTracker
Data type: bool
Access Specifier: ReadWrite

STARTRACKER GREEN MARKER GAIN

Path: **output/network/hidden-markers/startracker/green/marker-gain**
Description: Gets or sets the current brightness of the green StarTracker markers as a percentage
Data type: int
Units: Percentage
Range: 0 - 100
Access Specifier: ReadWrite

STARTRACKER GREEN MARKER SEED

Path: **output/network/hidden-markers/startracker/green/marker-seed**
Description: Gets or sets seed value for green StarTracker marker positions
Data type: int
Range: 1 - 65535
Access Specifier: ReadWrite

STARTRACKER GREEN MARKER SIZE SCALER

Path: **output/network/hidden-markers/startracker/green/marker-size-scaler**
Description: Gets or sets the size scaling for green StarTracker markers as a percentage
Data type: int
Units: Percentage
Range: 10 - 200
Access Specifier: ReadWrite

STARTRACKER RED DISTANCE TO TRACKER

Path: **output/network/hidden-markers/startracker/red/distance-to-tracker**
Description: Gets or sets distance to tracker for red StarTracker markers in metres
Data type: float

Units: Metres
Range: 0.1 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

STARTRACKER RED ENABLED

Path: **output/network/hidden-markers/startracker/red/enabled**

Description: Enables or disables red markers for StarTracker

Data type: bool

Access Specifier: ReadWrite

STARTRACKER RED MARKER GAIN

Path: **output/network/hidden-markers/startracker/red/marker-gain**

Description: Gets or sets the current brightness of the red StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

STARTRACKER RED MARKER SEED

Path: **output/network/hidden-markers/startracker/red/marker-seed**

Description: Gets or sets seed value for red StarTracker marker positions

Data type: int

Range: 1 - 65535

Access Specifier: ReadWrite

STARTRACKER RED MARKER SIZE SCALER

Path: **output/network/hidden-markers/startracker/red/marker-size-scaler**

Description: Gets or sets the size scaling for red StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 10 - 200

Access Specifier: ReadWrite

SHUTTERSYNC CUSTOM FRAME RATE

Path: **output/network/shuttersync/angle-settings/custom-frame-rate**

Description: Gets or sets the value of the ShutterSync custom frame rate

Data type: float

Units: Frames per Second

Range: 1 - 250

Decimal places: 3

Access Specifier: ReadWrite

SHUTTERSYNC ANGLE

Path: **output/network/shuttersync/angle-settings/shutter-angle**

Description: Gets or sets the value of the shutter angle

Data type: float

Units: Degrees

Range: 1 - 360

Decimal places: 3

Access Specifier: ReadWrite

SHUTTERSYNC USE CUSTOM FRAME RATE

Path: **output/network/shuttersync/angle-settings/use-custom-frame-rate**

Description: Enables or disables ShutterSync Custom Frame Rate

Data type: bool

Access Specifier: ReadWrite

SHUTTERSYNC DARK TIME

Path: **output/network/shuttersync/dark-time**

Description: Gets or sets the value of the ShutterSync dark time in milliseconds. Value stored independently of percentage value.

Data type: float

Units: Milliseconds

Range: 0 - 100

Decimal places: 3

Access Specifier: ReadWrite

SHUTTERSYNC DARK TIME MODE

Path: **output/network/shuttersync/dark-time-mode**

Description: Gets or sets the ShutterSync dark time mode. The processor can hold the dark time mode as a percentage and a time in milliseconds. This endpoint selects which value the processor uses.

Data type: enum

Supported values: time, percentage

Access Specifier: ReadWrite

SHUTTERSYNC DARK TIME PERCENTAGE

Path: **output/network/shuttersync/dark-time-percentage**

Description: Gets or sets the value of the ShutterSync dark time as a percentage of a frame. Value stored independently of milliseconds value.

Data type: float

Units: Percentage

Range: 0 - 100

Decimal places: 1

Access Specifier: ReadWrite

SHUTTERSYNC INSERT DARK TIME

Path: **output/network/shuttersync/insert-dark-time**

Description: Enables or disables insertion of dark time. Inserting dark time corresponds to displaying black frames in-between camera exposures.

Data type: bool

Access Specifier: ReadWrite

SHUTTERSYNC MODE

Path: **output/network/shuttersync/mode**

Description: Gets or sets ShutterSync operation mode

Data type: enum

Supported values: none, speed, angle

Access Specifier: ReadWrite

SHUTTERSYNC PRIORITISE REFRESH RATE

Path: **output/network/shuttersync/prioritise-refresh-rate**

Description: Enables or disables ShutterSync prioritise refresh rate setting

Data type: bool

Access Specifier: ReadWrite

SHUTTERSYNC SENSOR READOUT TIME

Path: **output/network/shuttersync/sensor-readout-time**

Description: Gets or sets sensor readout time

Data type: float

Units: Milliseconds

Range: 0.00 - 42.00

Decimal places: 2
Access Specifier: ReadWrite

SHUTTERSYNC SENSOR TYPE

Path: **output/network/shuttersync/sensor-type**
Description: Gets or sets ShutterSync sensor type
Data type: enum
Supported values: any, global-shutter, rolling-shutter
Access Specifier: ReadWrite

SHUTTERSYNC SPEED

Path: **output/network/shuttersync/speed-settings/shutter-speed**
Description: Gets or sets the denominator value of ShutterSync speed, numerator is 1
Data type: float
Range: 10 - 250
Decimal places: 3
Access Specifier: ReadWrite

SHUTTERSYNC TIME

Path: **output/network/shuttersync/speed-settings/time**
Description: Gets or sets the time value of ShutterSync time
Data type: float
Units: Milliseconds
Range: 4 - 100
Decimal places: 3
Access Specifier: ReadWrite

SHUTTERSYNC VIEWER

Path: **output/network/shuttersync/viewer**
Description: Gets or sets the ShutterSync viewer mode
Data type: enum
Supported values: eye, camera
Access Specifier: ReadWrite

OLD STARTRACKER BACKGROUND GAIN

Path: **output/network/startracker/background-gain**
Description: Gets or sets brightness of the video underneath StarTracker markers as a percentage
Data type: int
Units: Percentage
Range: 0 - 100
Access Specifier: ReadWrite

OLD STARTRACKER BLACKOUT AFFECTS MARKERS

Path: **output/network/startracker/blackout-affects-markers**
Description: Gets or sets whether blackout affects markers as well as video
Data type: bool
Access Specifier: ReadWrite

OLD STARTRACKER BLUE DISTANCE TO TRACKER

Path: **output/network/startracker/blue/distance-to-tracker**
Description: Gets or sets distance to tracker for blue StarTracker markers in metres
Data type: float
Units: Metres
Range: 0.1 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

OLD STARTRACKER BLUE ENABLED

Path: **output/network/startracker/blue/enabled**

Description: Enables or disables blue markers for StarTracker

Data type: bool

Access Specifier: ReadWrite

OLD STARTRACKER BLUE MARKER GAIN

Path: **output/network/startracker/blue/marker-gain**

Description: Gets or sets the current brightness of the blue StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

OLD STARTRACKER BLUE MARKER SIZE SCALER

Path: **output/network/startracker/blue/marker-size-scaler**

Description: Gets or sets the size scaling for blue StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 10 - 200

Access Specifier: ReadWrite

OLD STARTRACKER BLUE MARKER SEED

Path: **output/network/startracker/blue/star-map-seed**

Description: Gets or sets seed value for blue StarTracker marker positions

Data type: int

Range: 1 - 65535

Access Specifier: ReadWrite

OLD STARTRACKER ENABLED

Path: **output/network/startracker/enabled**

Description: Enables or disables StarTracker markers

Data type: bool

Access Specifier: ReadWrite

OLD STARTRACKER FRAMES ENABLED ON

Path: **output/network/startracker/frames-enabled-on**

Description: A valid array must contain ints in the range between 1 and the max supported frame rate multiplier for the platform

Data type: array

Access Specifier: ReadWrite

OLD STARTRACKER GREEN DISTANCE TO TRACKER

Path: **output/network/startracker/green/distance-to-tracker**

Description: Gets or sets distance to tracker for green StarTracker markers in metres

Data type: float

Units: Metres

Range: 0.1 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

OLD STARTRACKER GREEN ENABLED

Path: **output/network/startracker/green/enabled**

Description: Enables or disables green markers for StarTracker

Data type: bool

Access Specifier: ReadWrite

OLD STARTRACKER GREEN MARKER GAIN

Path: **output/network/startracker/green/marker-gain**

Description: Gets or sets the current brightness of the green StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

OLD STARTRACKER GREEN MARKER SIZE SCALER

Path: **output/network/startracker/green/marker-size-scaler**

Description: Gets or sets the size scaling for green StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 10 - 200

Access Specifier: ReadWrite

OLD STARTRACKER GREEN MARKER SEED

Path: **output/network/startracker/green/star-map-seed**

Description: Gets or sets seed value for green StarTracker marker positions

Data type: int

Range: 1 - 65535

Access Specifier: ReadWrite

OLD STARTRACKER RED DISTANCE TO TRACKER

Path: **output/network/startracker/red/distance-to-tracker**

Description: Gets or sets distance to tracker for red StarTracker markers in metres

Data type: float

Units: Metres

Range: 0.1 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

OLD STARTRACKER RED ENABLED

Path: **output/network/startracker/red/enabled**

Description: Enables or disables red markers for StarTracker

Data type: bool

Access Specifier: ReadWrite

OLD STARTRACKER RED MARKER GAIN

Path: **output/network/startracker/red/marker-gain**

Description: Gets or sets the current brightness of the red StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 0 - 100

Access Specifier: ReadWrite

OLD STARTRACKER RED MARKER SIZE SCALER

Path: **output/network/startracker/red/marker-size-scaler**

Description: Gets or sets the size scaling for red StarTracker markers as a percentage

Data type: int

Units: Percentage

Range: 10 - 200

Access Specifier: ReadWrite

OLD STARTRACKER RED MARKER SEED

Path: **output/network/startracker/red/star-map-seed**

Description: Gets or sets seed value for red StarTracker marker positions

Data type: int

Range: 1 - 65535

Access Specifier: ReadWrite

OVERRIDE

BLACKOUT ENABLED

Path: **override/blackout/enabled**

Description: Enables or disables blackout

Data type: bool

Access Specifier: ReadWrite

BLACKOUT FADE TIME

Path: **override/blackout/fade-time**

Description: The value of the blackout fade time. The fade time may be adjusted between zero (snap) and 10 seconds

Data type: float

Units: Seconds

Range: 0.0 - 10.0

Decimal places: 1

Access Specifier: ReadWrite

FREEZE ENABLED

Path: **override/freeze/enabled**

Description: Enables or disables video freeze

Data type: bool

Access Specifier: ReadWrite

TEST PATTERN CUSTOM COLOUR BLUE

Path: **override/test-pattern/custom-colour/blue**

Description: Gets or sets the custom colour test pattern blue value

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM COLOUR GREEN

Path: **override/test-pattern/custom-colour/green**

Description: Gets or sets the custom colour test pattern green value

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM COLOUR RED

Path: **override/test-pattern/custom-colour/red**

Description: Gets or sets the custom colour test pattern red value

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT END BLUE

Path: **override/test-pattern/custom-gradient/end-colour/blue**

Description: Gets or sets blue component of custom gradient test pattern end colour as a 12 bit integer

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT END GREEN

Path: **override/test-pattern/custom-gradient/end-colour/green**

Description: Gets or sets green component of custom gradient test pattern end colour as a 12 bit integer

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT END RED

Path: **override/test-pattern/custom-gradient/end-colour/red**

Description: Gets or sets red component of custom gradient test pattern end colour as a 12 bit integer

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT ORIENTATION

Path: **override/test-pattern/custom-gradient/orientation**

Description: Gets or sets the custom gradient test pattern orientation

Data type: enum

Supported values: horizontal, vertical

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT START GREEN

Path: **override/test-pattern/custom-gradient/start-colour/blue**

Description: Gets or sets green component of custom gradient test pattern start colour as a 12 bit integer

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT START BLUE

Path: **override/test-pattern/custom-gradient/start-colour/green**

Description: Gets or sets blue component of custom gradient test pattern start colour as a 12 bit integer

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN CUSTOM GRADIENT START RED

Path: **override/test-pattern/custom-gradient/start-colour/red**

Description: Gets or sets red component of custom gradient test pattern start colour as a 12 bit integer

Data type: int

Range: 0 - 4095

Access Specifier: ReadWrite

TEST PATTERN ENABLED

Path: **override/test-pattern/enabled**

Description: Enables or disables test pattern output function

Data type: bool

Access Specifier: ReadWrite

TEST PATTERN FORMAT

Path: **override/test-pattern/format**

Description: Format of the generated test pattern

Data type: enum

Supported values: from-input, standard-dynamic-range, perceptual-quantiser, hybrid-log-gamma

Access Specifier: ReadWrite

CAPTURE FRAME

Path: **override/test-pattern/frame-store/capture-frame**

Description: Captures the current frame and saves it to the frame store with the user number provided.

(Warning: Specifying a user number that already exists will overwrite the existing frame. This operation cannot be undone.)

Data type: int

Range: 1 - 50

Access Specifier: WriteOnly

DELETE FRAME

Path: **override/test-pattern/frame-store/delete-frame**

Description: Delete the frame store frame at the user number provided.

Data type: int

Range: 1 - 50

Access Specifier: WriteOnly

FRAME STORE COLOUR SPACE

Path: **override/test-pattern/frame-store/frames/{frame-user-number}/colour-space**

Description: Set colour space for the frame

Data type: enum

Supported values: rec-2020, dci-p3, rec-709, aces-cg, custom

Access Specifier: WriteOnly

FRAME STORE ALPHA ENABLED

Path: **override/test-pattern/frame-store/frames/{frame-user-number}/enable-alpha**

Description: Set alpha mode for the frame

Data type: bool

Access Specifier: ReadWrite

FRAME STORE FORMAT

Path: **override/test-pattern/frame-store/frames/{frame-user-number}/format**

Description: Set format for the frame

Data type: enum

Supported values: from-input, standard-dynamic-range, perceptual-quantiser, hybrid-log-gamma

Access Specifier: WriteOnly

FRAME NAME

Path: **override/test-pattern/frame-store/frames/{frame-user-number}/name**

Description: Name of the Frame

Data type: string

Access Specifier: ReadWrite

FRAME STORE SCALING MODE

Path: **override/test-pattern/frame-store/frames/{frame-user-number}/scaling-mode**

Description: Set scaling mode for the frame

Data type: enum

Supported values: 1:1, stretch, fit, fill

Access Specifier: WriteOnly

TEST PATTERN RESTRICT TO ACHIEVABLE COLOURS

Path: **override/test-pattern/restrict-to-achievable-colours**

Description: Enables or disables restrict to achievable colours switch

Data type: bool

Access Specifier: ReadWrite

TEST PATTERN TYPE

Path: **override/test-pattern/type**

Description: Determines which test pattern or frame store image will be shown when the test pattern widget is activated. Accepts either a frame store user number, or test pattern title. Defaults to SMPTE bars.

Data type: testpatterntype

Range: 1 - 50

Supported values: brompton, red, green, blue, cyan, magenta, yellow, white, black, grid, scrolling-grid, checkerboard, scrolling-checkerboard, colour-bars, scrolling-colour-bars, gradient, scrolling-gradient, strobe, smpte-bars, scrolling-smpte-bars, custom-colour, custom, forty-five-degree-grid, scrolling-forty-five-degree-grid, custom-gradient, scrolling-custom-gradient

Access Specifier: ReadWrite

PRESETS

ACTIVE PRESET NAME

Path: **presets/active/name**

Description: Name of the currently active preset

Data type: string

Access Specifier: ReadOnly

ACTIVE PRESET NUMBER

Path: **presets/active/number**

Description: Set to activate a preset

Data type: int

Range: 1 - 128

Access Specifier: ReadWrite

PRESET NAME

Path: **presets/items/{number}/name**

Description: Processor preset name

Data type: string

Access Specifier: ReadWrite

PRESET STATUS

Path: **presets/items/{number}/status**

Description: Preset activation status

Data type: bool

Access Specifier: ReadOnly

PROCESSING

3D LUT DATA

Path: **processing/3d-lut/data**

Description: Send bulk data in .cube file format to upload 3D LUT

Data type: bytearray

Access Specifier: ReadWrite

3D LUT ENABLED

Path: **processing/3d-lut/enabled**

Description: Enables or disables 3D LUT mapping

Data type: bool

Access Specifier: ReadWrite

3D LUT FILENAME

Path: **processing/3d-lut/filename**

Description: The name of the stored 3D LUT file

Data type: string

Access Specifier: ReadOnly

3D LUT STRENGTH

Path: **processing/3d-lut/strength**

Description: Intensity/opacity of 3D LUT

Data type: float

Units: Percentage

Range: 0.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

14-WAY COLOUR CORRECT BLACK BLUE

Path: **processing/colour-correct/black/blue**

Description: Gets or sets the value of the black blue

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

14-WAY COLOUR CORRECT BLACK GREEN

Path: **processing/colour-correct/black/green**

Description: Gets or sets the value of the black green

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

14-WAY COLOUR CORRECT BLACK RED

Path: **processing/colour-correct/black/red**

Description: Gets or sets the value of the black red

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

14-WAY COLOUR CORRECT BLUE BRIGHTNESS

Path: **processing/colour-correct/blue/brightness**

Description: Gets or sets the value of the blue brightness

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT BLUE HUE

Path: **processing/colour-correct/blue/hue**

Description: Gets or sets the value of the blue hue

Data type: float

Units: Degrees

Range: -180.0 - 180.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT BLUE SATURATION

Path: **processing/colour-correct/blue/saturation**

Description: Gets or sets the value of the blue saturation

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT COBALT BRIGHTNESS

Path: **processing/colour-correct/cobalt/brightness**

Description: Gets or sets the value of the cobalt brightness

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT COBALT HUE

Path: **processing/colour-correct/cobalt/hue**

Description: Gets or sets the value of the cobalt hue

Data type: float

Units: Degrees

Range: -180.0 - 180.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT COBALT SATURATION

Path: **processing/colour-correct/cobalt/saturation**

Description: Gets or sets the value of the cobalt saturation

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT CRIMSON BRIGHTNESS

Path: **processing/colour-correct/crimson/brightness**

Description: Gets or sets the value of the crimson brightness

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT CRIMSON HUE

Path: **processing/colour-correct/crimson/hue**

Description: Gets or sets the value of the crimson hue
Data type: float
Units: Degrees
Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT CRIMSON SATURATION

Path: **processing/colour-correct/crimson/saturation**
Description: Gets or sets the value of the crimson saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT CYAN BRIGHTNESS

Path: **processing/colour-correct/cyan/brightness**
Description: Gets or sets the value of the cyan brightness
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT CYAN HUE

Path: **processing/colour-correct/cyan/hue**
Description: Gets or sets the value of the cyan hue
Data type: float
Units: Degrees
Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT CYAN SATURATION

Path: **processing/colour-correct/cyan/saturation**
Description: Gets or sets the value of the cyan saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT ENABLED

Path: **processing/colour-correct/enabled**
Description: Enables or disables the processor's 14-Way Colour Correct feature
Data type: bool
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT GREEN BRIGHTNESS

Path: **processing/colour-correct/green/brightness**
Description: Gets or sets the value of the green brightness
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT GREEN HUE

Path: **processing/colour-correct/green/hue**

Description: Gets or sets the value of the green hue

Data type: float

Units: Degrees

Range: -180.0 - 180.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT GREEN SATURATION

Path: **processing/colour-correct/green/saturation**

Description: Gets or sets the value of the green saturation

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT LIME BRIGHTNESS

Path: **processing/colour-correct/lime/brightness**

Description: Gets or sets the value of the lime brightness

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT LIME HUE

Path: **processing/colour-correct/lime/hue**

Description: Gets or sets the value of the lime hue

Data type: float

Units: Degrees

Range: -180.0 - 180.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT LIME SATURATION

Path: **processing/colour-correct/lime/saturation**

Description: Gets or sets the value of the lime saturation

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT MAGENTA BRIGHTNESS

Path: **processing/colour-correct/magenta/brightness**

Description: Gets or sets the value of the magenta brightness

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT MAGENTA HUE

Path: **processing/colour-correct/magenta/hue**

Description: Gets or sets the value of the magenta hue
Data type: float
Units: Degrees
Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT MAGENTA SATURATION

Path: **processing/colour-correct/magenta/saturation**
Description: Gets or sets the value of the magenta saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT ORANGE BRIGHTNESS

Path: **processing/colour-correct/orange/brightness**
Description: Gets or sets the value of the orange brightness
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT ORANGE HUE

Path: **processing/colour-correct/orange/hue**
Description: Gets or sets the value of the orange hue
Data type: float
Units: Degrees
Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT ORANGE SATURATION

Path: **processing/colour-correct/orange/saturation**
Description: Gets or sets the value of the orange saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT RED BRIGHTNESS

Path: **processing/colour-correct/red/brightness**
Description: Gets or sets the value of the red brightness
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT RED HUE

Path: **processing/colour-correct/red/hue**
Description: Gets or sets the value of the red hue
Data type: float
Units: Degrees

Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT RED SATURATION

Path: **processing/colour-correct/red/saturation**
Description: Gets or sets the value of the red saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT TURQUOISE BRIGHTNESS

Path: **processing/colour-correct/turquoise/brightness**
Description: Gets or sets the value of the turquoise brightness
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT TURQUOISE HUE

Path: **processing/colour-correct/turquoise/hue**
Description: Gets or sets the value of the turquoise hue
Data type: float
Units: Degrees
Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT TURQUOISE SATURATION

Path: **processing/colour-correct/turquoise/saturation**
Description: Gets or sets the value of the turquoise saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT VIOLET BRIGHTNESS

Path: **processing/colour-correct/violet/brightness**
Description: Gets or sets the value of the violet brightness
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT VIOLET HUE

Path: **processing/colour-correct/violet/hue**
Description: Gets or sets the value of the violet hue
Data type: float
Units: Degrees
Range: -180.0 - 180.0
Decimal places: 1
Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT VIOLET SATURATION

Path: **processing/colour-correct/violet/saturation**

Description: Gets or sets the value of the violet saturation

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT WHITE BLUE

Path: **processing/colour-correct/white/blue**

Description: Gets or sets the value of the white blue

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT WHITE GREEN

Path: **processing/colour-correct/white/green**

Description: Gets or sets the value of the white green

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT WHITE RED

Path: **processing/colour-correct/white/red**

Description: Gets or sets the value of the white red

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT YELLOW BRIGHTNESS

Path: **processing/colour-correct/yellow/brightness**

Description: Gets or sets the value of the yellow brightness

Data type: float

Units: Percentage

Range: -100.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT YELLOW HUE

Path: **processing/colour-correct/yellow/hue**

Description: Gets or sets the value of the yellow hue

Data type: float

Units: Degrees

Range: -180.0 - 180.0

Decimal places: 1

Access Specifier: ReadWrite

I4-WAY COLOUR CORRECT YELLOW SATURATION

Path: **processing/colour-correct/yellow/saturation**

Description: Gets or sets the value of the yellow saturation
Data type: float
Units: Percentage
Range: -100.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

COLOUR REPLACE APPLY TO BRIGHTNESS

Path: **processing/colour-replace/apply-to-brightness**
Description: Enables or disables apply to brightness switch for colour replace
Data type: bool
Access Specifier: ReadWrite

COLOUR REPLACE APPLY TO HUE

Path: **processing/colour-replace/apply-to-hue**
Description: Enables or disables apply to hue switch for colour replace
Data type: bool
Access Specifier: ReadWrite

COLOUR REPLACE APPLY TO SATURATION

Path: **processing/colour-replace/apply-to-saturation**
Description: Enables or disables apply to saturation switch for colour replace
Data type: bool
Access Specifier: ReadWrite

COLOUR REPLACE BRIGHTNESS TOLERANCE

Path: **processing/colour-replace/brightness-tolerance**
Description: Gets or sets the value of the brightness tolerance for colour replace
Data type: float
Units: Percentage
Range: 10.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

COLOUR REPLACE FROM BLUE

Path: **processing/colour-replace/colour-from/blue**
Description: Gets or sets the blue component of the colour replace source colour
Data type: int
Range: 0 - 255
Access Specifier: ReadWrite

COLOUR REPLACE FROM GREEN

Path: **processing/colour-replace/colour-from/green**
Description: Gets or sets the green component of the colour replace source colour
Data type: int
Range: 0 - 255
Access Specifier: ReadWrite

COLOUR REPLACE FROM RED

Path: **processing/colour-replace/colour-from/red**
Description: Gets or sets the red component of the colour replace source colour
Data type: int
Range: 0 - 255
Access Specifier: ReadWrite

COLOUR REPLACE TO BLUE

Path: **processing/colour-replace/colour-to/blue**

Description: Gets or sets the blue component of the colour replace target colour

Data type: int

Range: 0 - 255

Access Specifier: ReadWrite

COLOUR REPLACE TO GREEN

Path: **processing/colour-replace/colour-to/green**

Description: Gets or sets the green component of the colour replace target colour

Data type: int

Range: 0 - 255

Access Specifier: ReadWrite

COLOUR REPLACE TO RED

Path: **processing/colour-replace/colour-to/red**

Description: Gets or sets the red component of the colour replace target colour

Data type: int

Range: 0 - 255

Access Specifier: ReadWrite

COLOUR REPLACE COLOUR TOLERANCE

Path: **processing/colour-replace/colour-tolerance**

Description: Gets or sets the value of the colour tolerance for colour replace

Data type: float

Units: Percentage

Range: 10.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

COLOUR REPLACE ENABLED

Path: **processing/colour-replace/enabled**

Description: Enables or disables the processor's Colour Replace feature

Data type: bool

Access Specifier: ReadWrite

COLOUR REPLACE METHOD

Path: **processing/colour-replace/method**

Description: Gets or sets the method applied by the processor for colour replacement

Data type: enum

Supported values: set-to-colour, transform-to-colour

Access Specifier: ReadWrite

COLOUR REPLACE SOFTNESS

Path: **processing/colour-replace/softness**

Description: Gets or sets the value of the softness for colour replace

Data type: float

Units: Percentage

Range: 6.0 - 100.0

Decimal places: 1

Access Specifier: ReadWrite

COLOUR REPLACE STRENGTH

Path: **processing/colour-replace/strength**

Description: Gets or sets the value of the strength for colour replace

Data type: float

Units: Percentage

Range: 6.0 - 100.0
Decimal places: 1
Access Specifier: ReadWrite

COLOUR REPLACE VIEW MATTE

Path: **processing/colour-replace/view-matte**
Description: Enables or disables view matte for colour replace
Data type: bool
Access Specifier: ReadWrite

CURVES BLUE POINTS

Path: **processing/curves/blue/points**
Description: A valid array must contain JSON objects with x and y values specified in the range [0,1] e.g. [{x:0.2, y:0.2}, {x:0.64, y:0.77}], or send an empty array to clear all points. Minimum separation between points is 0.0625
Data type: array
Access Specifier: ReadWrite

CURVES ENABLED

Path: **processing/curves/enabled**
Description: Enables or disables the processor's Colour Curves feature
Data type: bool
Access Specifier: ReadWrite

CURVES GREEN POINTS

Path: **processing/curves/green/points**
Description: A valid array must contain JSON objects with x and y values specified in the range [0,1] e.g. [{x:0.2, y:0.2}, {x:0.64, y:0.77}], or send an empty array to clear all points. Minimum separation between points is 0.0625
Data type: array
Access Specifier: ReadWrite

CURVES RED POINTS

Path: **processing/curves/red/points**
Description: A valid array must contain JSON objects with x and y values specified in the range [0,1] e.g. [{x:0.2, y:0.2}, {x:0.64, y:0.77}], or send an empty array to clear all points. Minimum separation between points is 0.0625
Data type: array
Access Specifier: ReadWrite

CURVES WHITE POINTS

Path: **processing/curves/white/points**
Description: A valid array must contain JSON objects with x and y values specified in the range [0,1] e.g. [{x:0.2, y:0.2}, {x:0.64, y:0.77}], or send an empty array to clear all points. Minimum separation between points is 0.0625
Data type: array
Access Specifier: ReadWrite

OSCA MODULE CORRECTION ENABLED

Path: **processing/osca/module-correction-enabled**
Description: Enables or disables OSCA module correction
Data type: bool
Access Specifier: ReadWrite

OSCA SEAM CORRECTION ENABLED

Path: **processing/osca/seam-correction-enabled**

Description: Enables or disables OSCA seam correction

Data type: bool

Access Specifier: ReadWrite

SCALER

Path: **processing/scaler/enabled**

Description: Enables or disables scaler

Data type: bool

Access Specifier: ReadWrite

PROJECT

PROJECT NAME

Path: **project/name**

Description: Gets the name of the Project

Data type: string

Access Specifier: ReadOnly

SYSTEM

SYSTEM REBOOT

Path: **system/actions/reboot**

Description: If the processor password is set then this must be sent in the body. Otherwise send a blank string to trigger the reboot.

Data type: string

Access Specifier: WriteOnly

SYSTEM SHUTDOWN

Path: **system/actions/shutdown**

Description: If the processor password is set then this must be sent in the body. Otherwise send a blank string to trigger the shutdown.

Data type: string

Access Specifier: WriteOnly

CURRENT DATE AND TIME

Path: **system/current-date-time**

Description: Current date/time of processor in yyyy-MM-dd hh:mm:ss 24 hour format

Data type: string

Access Specifier: ReadOnly

CASE FAN I SPEED

Path: **system/fan/case/one/speed**

Description: Current speed of the Case Fan

Data type: float

Units: RPM

Range: 0 - 5000

Access Specifier: ReadOnly

CASE FAN I STATUS

Path: **system/fan/case/one/status**

Description: Current activation status of Case Fan 1
Data type: bool
Access Specifier: ReadOnly

CASE FAN 2 SPEED

Path: **system/fan/case/two/speed**
Description: Current speed of Case Fan 2
Data type: float
Units: RPM
Range: 0 - 5000
Access Specifier: ReadOnly

CASE FAN 2 STATUS

Path: **system/fan/case/two/status**
Description: Current activation status of Case Fan 2
Data type: bool
Access Specifier: ReadOnly

FPGA FAN SPEED

Path: **system/fan/fpga/speed**
Description: Current speed of the FPGA Fan. Supported only on SX40, S8.
Data type: float
Units: RPM
Range: 0 - 5000
Access Specifier: ReadOnly

FPGA FAN STATUS

Path: **system/fan/fpga/status**
Description: Current activation status of the FPGA Fan. Supported only on SX40, S8.
Data type: bool
Access Specifier: ReadOnly

PROCESSOR NAME

Path: **system/processor-name**
Description: The name of the processor
Data type: string
Access Specifier: ReadOnly

PROCESSOR TYPE

Path: **system/processor-type**
Description: Processor hardware model
Data type: enum
Supported values: m2, s4, s8, t1, t8, sx40
Access Specifier: ReadOnly

SERIAL NUMBER

Path: **system/serial-number**
Description: Gets the Serial Number of the processor
Data type: string
Access Specifier: ReadOnly

SOFTWARE VERSION

Path: **system/software-version**
Description: Current version of software in format x.y.z

Data type: string
Access Specifier: ReadOnly

AMBIENT TEMPERATURE

Path: **system/temperature/ambient**
Description: Current Ambient Temperature. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

CPU TEMPERATURE

Path: **system/temperature/cpu**
Description: Current temperature of the main CPU core. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

DSP TEMPERATURE

Path: **system/temperature/dsp**
Description: Current Temperature of the DSP. Supported by M2, T1, S4.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

ETHERNET COPPER PHY A TEMPERATURE

Path: **system/temperature/ethernet/copper/a**
Description: Current Temperature of Ethernet Copper Physical Interface A. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

ETHERNET COPPER PHY B TEMPERATURE

Path: **system/temperature/ethernet/copper/b**
Description: Current Temperature of Ethernet Copper Physical Interface B. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

ETHERNET SFP A TEMPERATURE

Path: **system/temperature/ethernet/sfp/a**
Description: Current Temperature of Ethernet SFP A. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

ETHERNET SFP B TEMPERATURE

Path: **system/temperature/ethernet/sfp/b**
Description: Current Temperature of Ethernet SFP B. Supported by SX40, S8.
Data type: float
Units: Celsius

Range: 0 - 200
Access Specifier: ReadOnly

ETHERNET SFP C TEMPERATURE

Path: **system/temperature/ethernet/sfp/c**
Description: Current Temperature of Ethernet SFP C. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

ETHERNET SFP D TEMPERATURE

Path: **system/temperature/ethernet/sfp/d**
Description: Current Temperature of Ethernet SFP D. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

FPGA TEMPERATURE

Path: **system/temperature/fpga**
Description: Current temperature of the FPGA core. Supported by SX40, S8, M2.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

FRONT TEMPERATURE

Path: **system/temperature/front**
Description: Current Temperature of the Front Panel. Supported by M2.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

GPU TEMPERATURE

Path: **system/temperature/gpu**
Description: Current Temperature of the GPU. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

MAIN BOARD TEMPERATURE

Path: **system/temperature/main**
Description: Current Temperature of the Main Board. Supported by SX40, S8.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

PSU TEMPERATURE

Path: **system/temperature/psu**
Description: Current Temperature of the PSU. Supported by SX40, S8.
Data type: float
Units: Celsius

Range: 0 - 200
Access Specifier: ReadOnly

REAR TEMPERATURE

Path: **system/temperature/rear**
Description: Current Temperature of the Rear Panel. Supported by M2.
Data type: float
Units: Celsius
Range: 0 - 200
Access Specifier: ReadOnly

UPTIME

Path: **system/uptime**
Description: Time since processor boot in DDd HHh MMm SSs format
Data type: string
Access Specifier: ReadOnly

4. PROTOCOL EXAMPLE USAGE

HTTP

IP Control functionality over HTTP on port 80 is accessed via the **/api** path root to distinguish it from other web services.

```
http://192.168.0.50/api/
```

All commands are accessible via the regular HTTP verbs GET (for read, list and help operations) and PUT (for set). There is also query-parameter based support for accessing all operations exclusively via GET for older clients that do not support extra verbs.

A successful HTTP request will return the value of the parameter that has been requested, including the new value if a write command is issued. For example: {"brightness":1000}.

An unsuccessful HTTP request will return the response code string and any error messages generated by the API, for example: {"response-code":"Path not found"}.

READING DATA

Use GET verb with target path. Example to get input video refresh rate:

```
Client:
GET http://192.168.0.50/api/input/active/refresh-rate
```

```
Server:
HTTP/1.1 200 OK
Content-Type: application/json
{ refresh-rate: 60 }
```

WRITING DATA

Use PUT verb including data to set in a "data" tag in a JSON body. Example to set output brightness:

```
Client:
PUT http://192.168.0.50/api/output/global-colour/brightness
Content-Type: application/json
{ "data": 5000 }
```

```
Server:
HTTP/1.1 200 OK
Content-Type: application/json
{ "brightness": 5000 }
```

Alternative using GET verb only: pass the value to set as a "set=" query parameter.

```
Client: GET http://192.168.0.50/api/output/global-colour/brightness?set=5000
```

HELP

Use GET verb passing "help=1" as a query parameter

```
Client:
  GET http://192.168.0.50/api/override?help=1

Server:
  HTTP/1.1 200 OK
  Content-Type: application/json
  {
    "override":{
      "blackout":{
        "enabled":{
          "Access Specifier":"R/W",
          "Details":"Enables or disables blackout",
          "Name":"Blackout Enabled",
          "Summary":"Enable blackout",
          "Type":"Boolean"
        },
        "fade-time":{
          "Access Specifier":"R/W",
          "Details":"The value of the blackout fade time. The fade time
may be adjusted between zero (snap) and 10 seconds",
          "Name":"Blackout Fade Time",
          "Summary":"Time taken to fade to black when blackout enabled",
          "Type":"Float (range: 0 - 10)"
        }
      },
      "freeze":{
        "enabled":{
          "Access Specifier":"R/W",
          "Details":"Enables or disables video freeze",
          "Name":"Freeze Enabled",
          "Summary":"Enable video freeze",
          "Type":"Boolean"
        }
      }
    }
  }
```

SETTING MULTIPLE ENDPOINTS

It is possible to query and set multiple endpoints using the same command. This method requires fewer calls to the API (is more efficient), and is generally advised when attempting to make a number of changes to the processor at the same time.

WITH GET REQUEST

It is possible to chain multiple commands in a basic GET request, by adding ?set=1 after the API path, and chaining multiple parameter/value pairs with the & symbol.

Example: to set a processor's active video input to the first SDI port, two commands are required.

```
http://192.168.0.50/api/input/active/source/port-type?set=sdi
http://192.168.0.50/api/input/active/source/port-number?set=1
```

We can combine these two commands into a single GET request.

```
http://192.168.0.50/api/input/active/source?set=1&port-type=sdi&port-number=1
```

WITH PUT REQUEST VIA JSON

Where a large number of parameters should be set at the same time, it may be preferable to use a PUT request, with the body of request containing the parameter/value pairs formatted as JSON.

Example: to set a processor's Output Brightness, Colour Temperature, and Output Colour Gains at the same time, we can format this into a single PUT request.

```
PUT http://192.168.0.50/api/output/global-colour
Content-type: application/JSON
{
  "data": {
    "brightness": 4000,
    "colour-temperature": 5600,
    "gains": {
      "blue": 200,
      "green": 200,
      "intensity": 200,
      "red": 97.8,
    }
  }
}
```

PASSING AN ARRAY OF VALUES INTO JSON

Some endpoints, such as specifying which frames to enable Hidden Markers on, require an array of values.

This requires careful formatting using correct JSON notation

Example: to enable Hidden Markers on frames 2 and 4 on a processor, the following PUT request can be used.

```
PUT http://192.168.0.50/api/output/network/hidden-markers/frames-enabled-or
Content-type: application/JSON
{
  "data" : {
    "frames-enabled-on": [
      2,
      4
    ]
  }
}
```

BASIC REQUESTS USING A WEB BROWSER

Basic HTTP GET requests can be sent using the URL address field of a web browser. This includes read, write, list and help operations. To read an API endpoint, enter the IP address of the processor, followed by the API path.

Example: `http://192.168.0.50/api/output/global-colour/brightness`



USING THE COMMAND LINE

It is possible to send API commands using the processor on both Windows and macOS. One recommended method of sending HTTP requests from the command line is using curl, which is a command line utility that is installed by default on both Windows and macOS. This can be accessed by running the Command Prompt program on Windows, or the Terminal application on macOS.

EXAMPLES

To make a simple HTTP request using curl, open your command line application, then type curl followed by the API path, as in the previous examples. You can also specify more complex requests using command line arguments.

Basic read example:

```
curl http://192.168.0.50/api/output/global-colour/brightness
```

Basic write example:

```
curl http://192.168.0.50/api/output/globalcolour/brightness?set=450
```

Writing data with PUT

```
curl -X PUT --header "Content-type:application/json" --data "{\"data\":450}"  
http://192.168.0.50/api/output/global-colour/brightness
```

Note the escaped quotes, \" \", around data and other text strings in --data .

Writing .cube file data

```
curl -X PUT --header "Content-Type:application/cube"  
--data-binary @test_3d_table.cube  
http://192.168.0.50/api/processing/3d-lut/data
```

USING BITFOCUS COMPANION

Bitfocus Companion can be used to control a range of 3rd-party devices over IP, using pre-written modules. A Tessera module is available for controlling Brompton processors, and contains a number of widely used actions and variables. This module uses the API to communicate with the processor over HTTP, but allows the user to set up their actions in a graphical format with Companion.

HTTP REQUESTS MODULE

To control features of the processor that are not covered by the Tessera module, it is possible to use the Generic: HTTP module found in Companion to send HTTP requests to the processor.

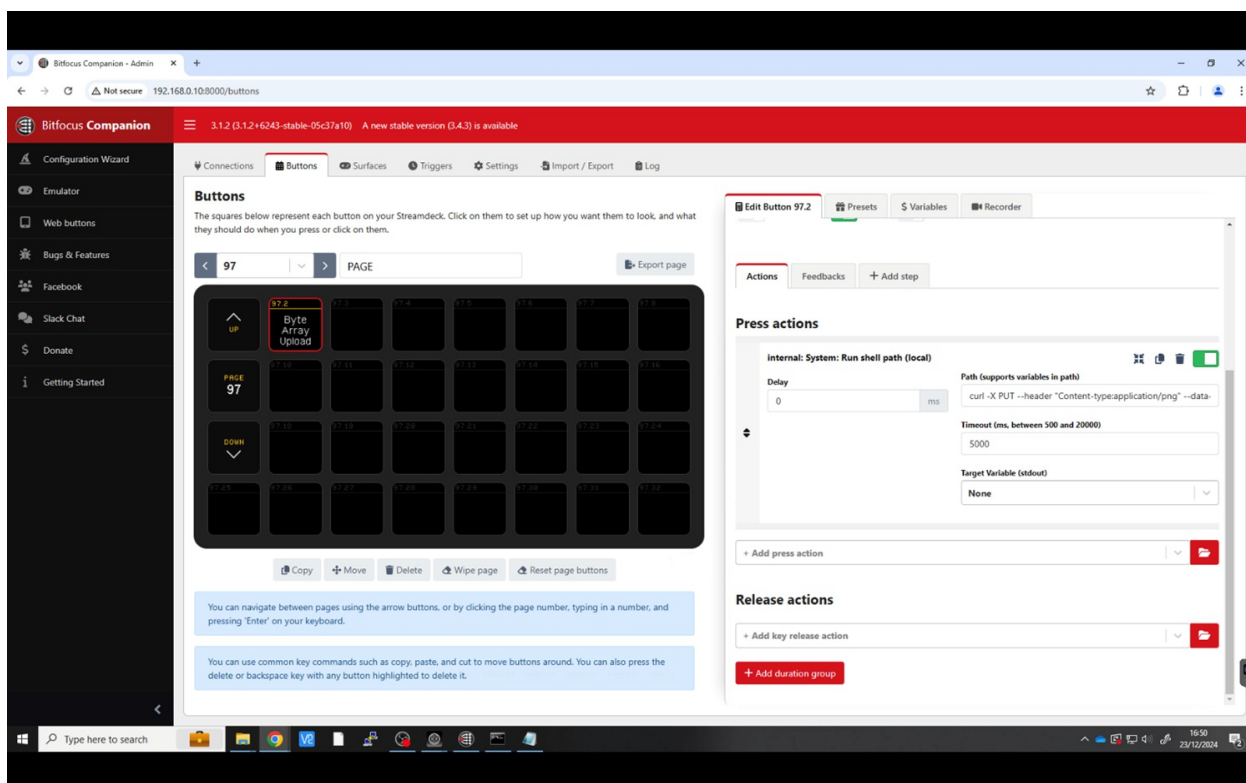
SYSTEM: RUN SHELL PATH

The **internal: System: Run shell path (local)** action can be used to run a terminal command from Companion. This can be useful in the case of uploading byte arrays (see Appendix A).

For example:

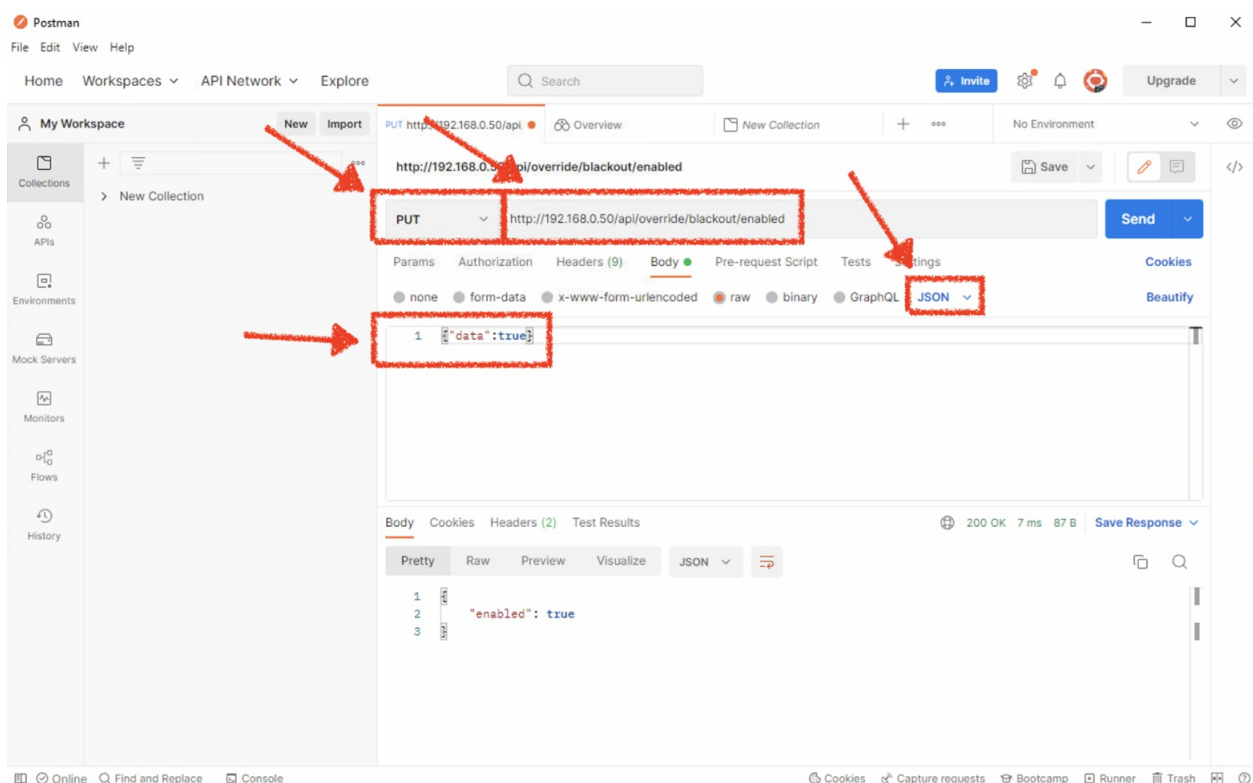
Adding the following path to the Path (supports variables in path) in a Run shell path action allows this PUT request to be triggered from Companion.

```
curl -X PUT --header "Content-type:application/png" --data-binary
@C:\Users\myUser\Documents\HiddenMarkers\Green.png
http://192.168.0.51/api/output/network/hidden-markers/custom/data
```



USING OTHER APPLICATIONS

Postman is web API client and testing platform, which offers a greater range of features for both sending and debugging requests. An example of a PUT request sent using Postman:



TELNET/TCP SOCKET

Telnet is a client/server protocol used for bi-directional communication across a network using TCP port 23. Telnet is case- and whitespace-sensitive, so commands will need to be entered exactly as written. If working inside a Telnet terminal window, delete and backspace often do not work, so careful typing is required. Alternatively, write the command out in full in a text editor, then copy and paste into your Telnet terminal.

WINDOWS

You may have to enable Telnet on Windows if you have not previously done so. One method of doing this is

- Open Command Prompt (right-click, Run as Administrator)
- `dism /online /Enable-Feature /FeatureName:TelnetClient`
- Close and re-open Command Prompt

To connect to a processor, open Command Prompt and enter `telnet 192.168.0.50 23`. In the new Telnet window that opens, you can send now Telnet commands to the processor. As noted above, it may be helpful to write out commands in a separate text editor. To paste a command into the Telnet terminal, right-click on the terminal window.

To set an endpoint: `set path/to/endpoint/n-1/ endpoint=value`

Example: `set output/global-colour/ brightness=1000`

When an endpoint has been set successfully, telnet will return the API tree from `/n-1/`.

To get an endpoint: `get path/to/endpoint/ .`

Example: `get output/global-colour/brightness` will return `brightness=1000 .`

MACOS

To use Telnet in the macOS terminal, Telnet must first be installed on the system. One recommended installation method is using homebrew package manager. If homebrew isn't already installed on the control device (e.g. laptop), see the homebrew documentation for installation instructions. To install Telnet, open the Terminal, and use the command `brew install telnet`. Once Telnet has been installed, follow the instructions for using Telnet on Windows to connect and control your processor.

5. APPENDICES

APPENDIX A | DATA TYPES EXAMPLE USAGE

Below are brief descriptions of each type, and some examples to show how they can be used.

STRING

Text string of up to 128 UTF-8 encoded characters.

Example endpoints: Device Type, Active Preset Name, Group Name, Frame Name.

GET request using web browser:

```
http://192.168.0.50/api/presets/active/name
```

PUT request using command line:

```
curl -X PUT --header "Content-type:application/json" --data
"{\"data\":\"Holding Card\"}"
http://192.168.0.50/api/override/test-pattern/frame-store/frames/1/name
```

BOOL(EAN)

Boolean state, true or false.

Example endpoints: Blackout Enabled, Freeze Enabled.

GET request using web browser:

```
http://192.168.0.50/api/override/blackout/enabled?set=true
```

PUT request using command line:

```
curl -X PUT --header "Content-type:application/json" --data
"{\"data\":true}"
http://192.168.0.50/api/override/blackout/enabled
```

INTEGER

Signed 16-bit integer, range -32768 to 32767.

Each endpoint has a range of accepted values; sending values outside this range will result in the request being rejected.

Example endpoints: Output Brightness, Output Colour Temperature.

GET request using web browser:

```
http://192.168.0.50/api/output/global-colour/brightness?set=450
```

PUT request using command line:

```
curl -X PUT --header "Content-type:application/json" --data
"{\"data\":450}"
http://192.168.0.50/api/output/global-colour/brightness
```

FLOAT

Floating point value (i.e. has a decimal place).

Each endpoint has a range of accepted values; sending values outside this range will result in the request being rejected.

Example endpoints: 3D LUT Strength, ShutterSync Angle, Genlock Phase Offset Fraction.

GET request using web browser:

```
http://192.168.0.50/api/processing/3d-lut/strength?set=33.3
```

PUT request using command line:

```
curl -X PUT --header "Content-type:application/json" --data
"{\"data\":33.3}"
http://192.168.0.50/api/processing/3d-lut/strength
```

BYTE ARRAY

This data type is used to send data to the 3D LUT Data and Hidden Markers Data.

It may be advisable to specify the HTTP request to point to a file containing the byte array data, rather than including the data itself in the HTTP request itself.

Example of sending a PNG image to the processor to be used as a hidden marker frame.

On Windows:

```
curl -X PUT --header "Content-type:application/png" --data-binary
@C:\Users\myUser\Documents\HiddenMarkers\Green.png
http://192.168.0.51/api/output/network/hidden-markers/custom/data
```

On macOS:

```
curl -X PUT --header "Content-type:application/png" --data-binary
@/Users/myUser/Documents/HiddenMarkers/Green.png
http://192.168.0.50/api/output/network/hidden-markers/custom/data
```

ENUM

String enumeration representing one of a discrete set of possible values.

Example endpoints: Genlock Source, Processor Type.

GET request using web browser:

```
http://192.168.0.50/api/output/network/genlock/source?set=sdi
```

PUT request using command line:

```
curl -X PUT --header "Content-type:application/json" --data
"{\"data\":\"sdi\"}"
http://192.168.0.50/api/output/network/genlock/source
```

TESTPATTERN TYPE

Either an enum representing a test pattern, or an integer representing a frame store image by its user number.

Used for Test Pattern Type endpoint. To set Test Pattern Type to one of the available test patterns, use one of the enums from the documentation (e.g. colour-bars). To recall a Frame Store, use the corresponding user number.

See notes and examples for Enum and Integer data types.